When Is Label-Free Adaptation Knowable? Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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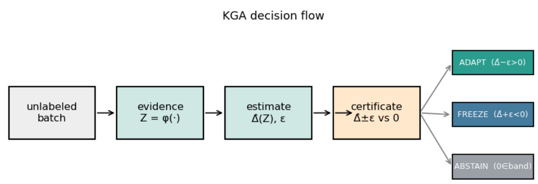

Fig. 1. **K-Bound previews the decision.** From an unlabeled batch under shift, the certificate estimates the adaptation benefit $\widehat{\Delta}$ with a finite-sample radius ε and returns ADAPT (knowably helpful), FREEZE (knowably harmful), or ABSTAIN (sign unidentifiable), with false-adapt/false-freeze $\leq \alpha$.

***Abstract*—Test-time adaptation (TTA) can improve a model on unlabeled target data, but the same update can also silently harm performance. We study the decision that comes *before* adaptation: should a model adapt, freeze, or abstain? We define the adaptation benefit as the target-risk difference between a frozen baseline and an adapted candidate. Our central result is an exact *benefit-sign frontier*: over a declared calibration-drift class, the sign of the adaptation benefit is identifiable from label-free evidence *if and only if* an observable disagreement-region margin exceeds the admitted drift budget, $|M| > \beta$. Below this frontier, two target worlds can produce identical label-free evidence while requiring opposite actions, making abstention necessary.**

We instantiate the framework as Knowability-Guided Adaptation (KGA), a mechanism-agnostic wrapper around existing TTA methods. Under stated coverage, exchangeability, and risk-alignment assumptions, KGA controls the unconditional false-adapt and false-freeze probabilities. Across held-out natural shifts, KGA is uniformly no-harm: it matches the better fixed policy and beats the worse. In identifiable mixed regimes, it beats fixed adapt/freeze policies and protocol-matched POEM-style and AETTA-style baselines; in a pre-registered pooled heterogeneous deployment, a single universal gate beats both fixed policies across seven natural sources. K-Bound is therefore a safety layer for adaptation, not a universal accuracy booster.

***Index Terms*—test-time adaptation, distribution shift, selective prediction, identifiability, label-free risk estimation**

I. Introduction

Test-time adaptation (TTA) updates a model on the unlabeled test stream and, under favorable distribution shift, recovers accuracy; under unfavorable shift the same objective silently degrades it. With no target labels, the system usually cannot tell which case it faces. The decision that matters therefore comes *before* the update: given a frozen baseline f_0 and an adapted candidate f_a, choose **adapt** when adaptation is knowably helpful, **freeze** when knowably harmful, or **abstain** when label-free evidence cannot identify the sign of the benefit.

Some cases are information-theoretically unknowable. That label-free quantities cannot decide adaptation without an assumption on the shift is established: identifying OOD accuracy is as hard as identifying the optimal predictor [13], and the optimal predict-versus-abstain rule under a shift budget has a known form [14]. We sharpen this landscape, for the benefit sign, into a computable criterion: whether adaptation helps or hurts is label-free identifiable *if and only if* an observable margin exceeds the calibration-drift budget on the disagreement region—$|M| > \beta$ (Theorem 2). On the unidentifiable side, two target worlds carry identical label-free evidence but opposite benefit, so no label-free rule can be correct in both (Theorem 1) and abstention is forced. KGA (Algorithm 1)—a benefit estimator plus a conformal radius, wrapped around any adapter (Tent, EATA, SAR)—operationalizes the frontier as a finite-sample certificate holding false-adapt and false-freeze at level α (Theorem 3).

The evidence follows the regime map the theory draws, summarized in Table I: held-out *natural* shifts are uniformly **no-harm** (KGA matches the better fixed policy and beats the worse; no single-dataset natural beats-both is claimed); *identifiable mixed* harmful+helpful regimes supply the **beats-both** policy wins, including a pre-registered head-to-head over protocol-matched POEM-style and AETTA-style baselines; and a pre-registered *pooled heterogeneous* deployment shows one universal gate beating both fixed policies across seven natural sources. K-Bound does *not* claim universal improvement: when adaptation is already safe, a valid certificate can only tie always-adapt. It is a safety layer for TTA, not an accuracy booster.

a) Contributions.:

TABLE I

Regime map: expected behavior vs. result. Each regime, an example dataset, the behavior the frontier predicts for a valid certificate, and what was measured. "No-harm" = matches the better fixed policy and beats the worse, without beating both; "beats both" = lower regret than always-adapt *and* always-freeze with CIs excluding zero. Per-dataset numbers: Tables II–IX; committal-gate baselines: Appendix M (Table XXIX).

Regime	Example	Expected	Result
Helpful-dominated	Camelyon17 OOD	tie always-adapt	no-harm
Harmful-dominated	iWildCam, RxRx1, Office-Home (aggr. SAR)	tie always-freeze, beat always-adapt	no-harm
Domain-gen. natural	PACS (4 splits)	no-harm	no-harm (ties)
Mixed + detectable	CIFAR-10-C stress, ImageNet-C SAR	beat both fixed policies	confirmed (CI-robust / operating point)
Weak evidence	ImageNet-R, CIFAR-10.1	abstain / conservative	diagnostic, not headline
Heterogeneous deployment	pooled seven-source stream	route per condition, beat both	universal-gate win (pre-reg.)

1) **Problem framing.** TTA as a label-free *adapt/freeze/abstain* decision—whether to adapt at all, not how to adapt—specializing the known impossibility landscape [13], [14] to the benefit sign.
2) **One exact result (the spine).** The *benefit-sign frontier*: $\operatorname{sign}\Delta$ is label-free identifiable over the declared drift class *if and only if* $|M| > \beta$ (Theorem 2); the "only if" is a matched-evidence impossibility (Theorem 1), the "if" yields a finite-sample certificate (Theorem 3). The proofs hold for any *split-observable* functional ($\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma)$, M observable, $|\gamma| \leq \beta$), with TTA the demonstrated instance.
3) **Unification.** ATC, DoC, GDE, COT, AETTA, and agreement-on-the-line are the zero-budget (β=0) face of the frontier—sound precisely when calibration does not drift.
4) **A drop-in method.** KGA, a mechanism-agnostic certificate wrapper around Tent/EATA/SAR (Algorithm 1).
5) **Regime-mapped evidence.** Uniform no-harm on five held-out natural benchmarks at $\mathrm{FA_u}$=0 (the primary deployment claim); beats-both plus a pre-registered win over protocol-matched POEM-style/AETTA-style baselines in identifiable mixed regimes; and a pre-registered universal-gate win on a pooled seven-source heterogeneous deployment (Table I).

Table note. Per-dataset quantitative results (regret, false-adapt, coverage) and the direct committal-gate baselines appear in the per-dataset tables and Appendix M.

II. Related work

Across every adjacent line, one distinction recurs—the one the theory formalizes: *improving* an unlabeled objective versus *certifying the sign* of its benefit before acting.

a) TTA methods.: Tent [1], SHOT [2], TTT [3], EATA [4], SAR [5], CoTTA [6], and MEMO [7] make adaptation more robust but commit to adapting; none certifies the sign of benefit *before* updating, and none abstains on the adapt/freeze decision itself.

b) Guarded and monitored TTA.: POEM [11] and sequential risk monitors [9], [10] act during or after adaptation and control false alarms in one failure direction. K-Bound decides *before* committing, poses a three-way decision with an explicit unknowable region, and controls false-*adapt* on the benefit sign.

c) Label-free accuracy and error estimation.: ATC [13], Agreement-on-the-Line [16], [15], and AETTA [18] estimate target performance without labels. Garg et al. [13] also establish the limit governing this whole line—identifying OOD accuracy is as hard as identifying the optimal predictor—so every such estimate is sound only under an assumption on the shift. We *organize* these estimators rather than compete with them: ATC, DoC, GDE, COT, AETTA, and agreement-on-the-line are all the zero-budget (β=0) face of our frontier, sound precisely when calibration does not drift. Most directly, [17] applies agreement-on-the-line to decide, at the *population* level, when TTA helps or hurts and to select an adaptation strategy—a plug-in use on that same β=0 face; it issues no per-deployment certificate, no ABSTAIN action, and no identifiability limit, and its linear trend is known to break on shifts (e.g. Gaussian-noise corruption)—precisely the regime where K-Bound abstains. Estimating $R_T(f)$ differs from certifying $\operatorname{sign}(R_T(f_0) - R_T(f_a))$ before committing the candidate update, with controlled false-adapt rate. Thus, K-Bound is not an accuracy estimator used as a gate; it changes the target estimand from target accuracy to the sign of the frozen-versus-adapted risk difference, with abstention when that sign is not identifiable.

d) Impossibility, abstention under shift, and identifiability.: The general impossibility we specialize is owned by prior work: Garg et al. [13] prove identifying OOD accuracy is as hard as identifying the optimal predictor, and Kalai & Kanade [14] give the matching decision theory—the optimal abstain-versus-predict rule under an observable-quantity-versus-shift-budget threshold, with a minimax certificate. We cite these as the sources and do not rederive them. Unsupervised risk estimation [21], [22], [25] and selective prediction [37] likewise show performance is identifiable only under structure. K-Bound *specializes* this landscape to the *sign* of adaptation benefit, certifying it before committing with an explicit abstain region and false-adapt control at level α. What none of these lines supplies—and what the next section makes exact—is the computable boundary ($|M| > \beta$) between the adaptation

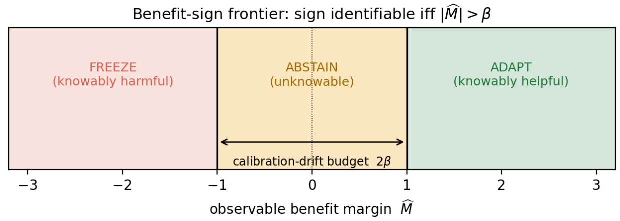

Fig. 2. **The benefit-sign frontier.** The sign of the adaptation benefit is identifiable from label-free evidence iff the observable margin $|\widehat{M}|$ exceeds the calibration-drift budget β. Outside the band KGA ADAPTS (knowably helpful) or FREEZES (knowably harmful); inside the band—the unknowable region—abstention is information-theoretically forced.

decisions one can certify label-free and those no rule ever can.

III. Problem setup and theory

Source $P_S(X, Y)$ has labels at train/validation; target $P_T(X, Y)$ exposes only $P_T(X)$ at test time. Fix loss ℓ, frozen f_0, and candidate f_a. Target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$. The *adaptation benefit* is

$$\Delta \;=\; R_T(f_0) - R_T(f_a), \qquad \Delta > 0 \iff \text{adapting helps.} \tag{1}$$

Observable evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable without target labels (entropy, confidence, drift, disagreement, update norms). The central question is when $\operatorname{sign}\Delta$ is recoverable from Z alone. The certificate needs two *separate* requirements. **(i) Risk-alignment** is a property of Z and the class: $\operatorname{sign}\Delta$ is a measurable function of the population law of Z over the declared class—equivalently, no two admissible worlds with opposite benefit sign induce the *same* Z-law. **(ii) Coverage** is a property of the calibrator: the benefit interval $[\widehat{\Delta} - \varepsilon, \widehat{\Delta} + \varepsilon]$ has marginal coverage $\geq 1 - \alpha$. Risk-alignment makes $\operatorname{sign}\Delta$ recoverable in principle; coverage makes the finite-sample certificate valid. When Z is *not* risk-aligned, no label-free rule can recover $\operatorname{sign}\Delta$ (Theorem 1) and the certificate must abstain.

What "label-free" means. No target/test labels at deploy; labeled source and held-out validation may calibrate $\widehat{\Delta}$ and ε. Evidence Z uses only unlabeled batches, predictions, entropy, drift, and update statistics. The method is **target-label-free**, not label-free of all supervision: labels may be used only in the source/validation calibration split, never in the deployment batch or held-out test decision.

K-Bound in one sentence. The true adaptation-benefit sign is recoverable only when label-free evidence exceeds the uncertainty created by calibration drift; otherwise the correct action is abstention.

a) Four quantities that must not be confused.:

Symbol	Meaning	At deploy?	Role
M	observable evidence margin	yes	evidence toward adapt/freeze
γ	realized calibration drift	no	can reverse the benefit sign
β	declared bound on $\lvert\gamma\rvert$	externally supplied	population frontier
ε	uncertainty radius from held-out residuals	estimated pre-deploy	finite-sample certificate

ε is an operational radius calibrated from residuals; it is *not* an estimate of the worst-case drift β.

Definition 1 (Regimes)**.** *Knowably helpful*: evidence certifies $\Delta > 0$ (adapt). *Knowably harmful*: evidence certifies $\Delta < 0$ (freeze). *Unknowable*: the same Z is compatible with both signs (abstain).

Theorem 1 (Matched evidence forces abstention)**.** *There exist worlds P_T^1, P_T^2 in the declared class with* $\operatorname{Law}(Z \mid P_T^1) = \operatorname{Law}(Z \mid P_T^2)$ *but* $\operatorname{sign}\Delta_1 = -\operatorname{sign}\Delta_2 \neq 0$. *No label-free rule can be correct in both; any rule with false-adapt and false-freeze each controlled at α must abstain on matched evidence with probability at least $1 - 2\alpha$.*

Intuition. Two target worlds can induce identical label-free evidence while adaptation helps in one and harms in the other; a rule that controls false adaptation and false freezing must abstain there. *Why it is true:* (1) a swap construction on the disagreement region produces two admissible worlds with the same evidence law and opposite benefit signs; (2) any label-free rule sees the same Z-law in both, so its action distribution is identical; (3) each committed action is a false commitment in one of the two worlds, so control at level α in both forces abstention with probability $\geq 1 - 2\alpha$. *Scope:* an impossibility over the declared class $\mathcal{C}_\beta$, not a statement about any single fixed target.

Theorem 2 (Exact benefit-sign frontier)**.** *On the disagreement region $D = \{x : f_0(x) \neq f_a(x)\}$, write $M = \mathbb{E}[s \mid D] - \frac{1}{2}$ for a source-calibrated score s of f_a and $\gamma = \mathbb{E}[\eta_a - s \mid D]$ for the* (unobserved) *calibration drift. Then*

$$\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma), \qquad |\gamma| \leq \beta,$$

and over the declared target class $\mathcal{C}_\beta = \{P_T : |\gamma| \leq \beta\}$ with supplied drift budget $\beta \geq 0$, the benefit sign is label-free identifiable if and only if

$$|M| > \beta.$$

Region	Meaning	Action
$M > \beta$	allowed drift cannot flip the positive sign	ADAPT
$M < -\beta$	allowed drift cannot flip the negative sign	FREEZE
$\lvert M\rvert \leq \beta$	drift can flip the sign	ABSTAIN

Intuition. The decision is safe exactly when the observable margin clears the admitted drift, and impossible when drift can dominate it. *Why it is true:* (1) the disagreement-region reduction confines the benefit to D and yields the split-observable decomposition $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$; (2) if $|M| > \beta$, every admissible drift leaves the sign of $M+\gamma$ equal to that of M, so the sign is a function of observables; (3) if $|M| \le \beta$, two admissible drifts realize opposite signs on matched evidence and Theorem 1 applies. *Scope:* population-level and exact in both directions; β must be externally justified for the deployment class (full dichotomy: Appendix A). *Worked example.* (A) M=0.18, β=0.10: $M-\beta = 0.08 > 0$, so adaptation stays beneficial under every allowed drift—ADAPT. (B) M=0.07, β=0.10: $M+\gamma$ may be negative or positive, so the sign is not identifiable—ABSTAIN.

Theorem 3 (Finite-sample adapt/freeze/abstain certificate)**.** *Suppose the benefit estimate satisfies coverage*

$$\mathbb{P}\Big[\,|\widehat{\Delta} - \Delta| \le \varepsilon\Big] \;\ge\; 1-\alpha.$$

Then the rule

$$g(Z) = \begin{cases} \textit{ADAPT} & \widehat{\Delta} - \varepsilon > 0, \\ \textit{FREEZE} & \widehat{\Delta} + \varepsilon < 0, \\ \textit{ABSTAIN} & \textit{otherwise}, \end{cases}$$

controls both marginal error probabilities:

$$\mathbb{P}[\textit{ADAPT},\, \Delta \le 0] \le \alpha, \qquad \mathbb{P}[\textit{FREEZE},\, \Delta \ge 0] \le \alpha.$$

Warning. This theorem controls the *marginal, unconditional* false-adapt and false-freeze probabilities ($\mathrm{FA_u}$). It does *not* bound the conditional false-adapt rate among committed ADAPT decisions, $\mathrm{FA_c} = \mathbb{P}[\Delta \le 0 \mid \text{ADAPT}]$, which is reported empirically.

Intuition. A valid uncertainty radius turns the population frontier into an operational rule: commit only when the entire benefit interval lies on one side of zero. *Why it is true:* (1) on the coverage event, $\widehat{\Delta} - \varepsilon > 0$ implies $\Delta > 0$; (2) a false adapt therefore requires the coverage event to fail; (3) that failure has probability at most α, and symmetrically for freeze. *Scope:* requires exchangeable calibration pairs, risk-aligned evidence, and bounded benefits (§IV); ε is an operational surrogate valid under these assumptions—not the frontier budget β.

Corollary 1 (Unjustified assumptions ⇒ abstain)**.** *The certificate's false-adapt guarantee holds only under the stated coverage/exchangeability and risk-alignment assumptions. When those assumptions cannot be justified for the deployment batch (e.g. the evidence is not risk-aligned, or the calibration sample is not exchangeable with deployment), the guarantee does not apply, and the conservative operational action is to abstain or freeze rather than commit to adaptation: committing adapt without a valid positive certified lower bound is not protected by the target false-adapt level α. (At deployment Δ is unobservable, so this is a statement about* assumptions*, not about observing $|\widehat{\Delta} - \Delta|$.)*

b) How the pieces fit.:

disagreement-region reduction → matched-evidence ambiguity → **Thm. 1** (abstention necessity)
→ split-observable decomposition $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ → **Thm. 2** ($|M| > \beta$ frontier)
→ coverage condition → **Thm. 3** (finite-sample certificate)

c) Extensions.: One-bit orientation—with the weakest one-bit class characterized unconditionally as a finite family of dominance polytopes (Appendix P, Theorem 10)—minimax order-optimality on the identifiable side, multicandidate family-wise routing, anytime-valid streaming, and the Wave-4 dichotomies (exact three-world minimax constant, margin computability, multiclass capacity, regression bracketing) are proved and validated in Appendix A and the long manuscript (`kbound.tex`). They strengthen the framework but are not required to understand the main K-Bound result. The streaming form is demonstrated on real data: on the native 35,370-image iWildCam stream (45 windows, 75.6% harmful), the betting e-process detects continual-Tent collapse at window 6 with anytime false-adapt 0, and the resulting gate value (0.6995) ties the oracle policy exactly (pre-registered).

IV. Method: KGA

KGA turns the frontier certificate into a deployable rule. It is the only headline method—a decision wrapper around a fixed adapter, not a new adaptation objective. We fit the benefit estimator and its radius *out of fold*, so the data used to fit $\widehat{\Delta}$ and the data used to calibrate its residual are disjoint for every point. On the per-cell stress grids (CIFAR-10-C and the synthetic grids): for each cell i a gradient-boosted $\widehat{\Delta}_{-i}$ is trained on the other $N-1$ cells and evaluated on cell i, and ε is the $(1-\alpha)$ empirical quantile of the leave-one-out residuals $\{|\widehat{\Delta}_{-i}(Z_i) - \Delta_i|\}_{i=1}^{N}$—leave-one-out (jackknife) cross-conformal, so no cell's residual uses an estimator that saw it. On the natural-shift protocols (Camelyon17, iWildCam, Office-Home) the split is across domains: $\widehat{\Delta}$ and ε are fit once on a dev split and the held-out test domain is scored a single time—ordinary split-conformal. At deploy we extract Z from an unlabeled batch and return adapt, freeze, or abstain (Algorithm 1). Labels enter only through the calibration split; the deployment batch and held-out test scorer never use target labels for the adapter, ε, evidence, or rule selection. The coverage guarantee is split-dependent. The natural-shift protocols use a clean dev/test split: *split-conformal* coverage $\mathbb{P}[|\widehat{\Delta} - \Delta| \le \varepsilon] \ge 1-\alpha$, exact in finite samples under exchangeability of the calibration and test cells. The per-cell grids set ε to the $(1-\alpha)$ empirical quantile of the leave-one-out residuals, so the *calibration-set* coverage is $\ge 1-\alpha$ by construction (realized 0.898 at nominal 0.90, α=0.10)—the standard jackknife radius, which is distribution-free only asymptotically under estimator stability; its formal finite-sample counterpart is the jackknife+ of Barber et al. (2021) at level $1-2\alpha$, which we note but do not implement. The decision rule's false-

Algorithm 1 KGA: Knowability-Guided Adaptation

Require: Frozen f_0; adapter $\mathcal{A}$; batch $X_{1:m}$; $\widehat{\Delta}$, ε, α
Ensure: Decision $g \in \{\textsc{Adapt}, \textsc{Freeze}, \textsc{Abstain}\}$

```
f_a ← A(f_0, X_{1:m})
Z ← φ(X_{1:m}, f_0, f_a)
Δ̂ ← Δ̂(Z);  L ← Δ̂ − ε;  U ← Δ̂ + ε
if L > 0 then
    g ← ADAPT
else if U < 0 then
    g ← FREEZE
else
    g ← ABSTAIN
end if
return g
```

adapt guarantee is stated under the coverage hypothesis (§IV), which the empirical coverage meets.

On the good event $|\widehat{\Delta} - \Delta| \le \varepsilon$, committing adapt only when $L > 0$ implies $\Delta > 0$ (Theorem 3). Assumptions: exchangeable calibration pairs, bounded benefits, and risk-aligned Z; no target labels at deploy. Overhead: one forward pass of f_0 and f_a plus lightweight $\widehat{\Delta}$ evaluation.

a) The threshold is derived, not tuned.: KGA's commit boundary is not a swept hyperparameter, but the population frontier and the empirical radius must be kept distinct. The *population* frontier of Theorem 2—over the declared class $\mathcal{C}_\beta$, sign Δ is identifiable iff the disagreement margin $|M|$ exceeds the supplied budget β—fixes the *required* budget at population level; the drift realization $\gamma \le \beta$ is unknown and not observed at deploy. KGA does not observe γ; it commits under a *conservative empirical radius* ε, the $(1-\alpha)$ residual quantile on the calibration split, fixed there and *never* selected on target-test data. The two are related but not identical: ε is a finite-sample, assumption-dependent surrogate that controls false-adapt under the stated exchangeability and risk-alignment assumptions—not the frontier itself. Confidence-thresholding (ATC) and agreement-trend selection (AGL) fit a scalar against a labeled or held-out signal and predict *accuracy*; KGA certifies the benefit *sign* with a boundary *forced* by the identifiability budget, and—unlike any thresholded estimator—ABSTAINS on the region where Theorem 1 *proves* no label-free rule can decide. The impossibility construction, not a tuned cutoff, is what separates KGA from "estimate accuracy, then threshold."

V. Experiments

The experiments test one claim: KGA is a safety layer, beating both fixed policies only where harmful adaptation is frequent and detectable, and doing no harm everywhere else. We evaluate it on two controlled corruption/stress tracks (CIFAR-10-C stress, ImageNet-C SAR) and three held-out natural-shift replications (Protocols G, H v2, and M v2; Table I). Calibration and held-out scoring follow Algorithm 2 (Appendix A). Metrics: mean accuracy; regret-to-oracle against the per-condition oracle $g^\star = \mathbf{1}[\Delta > 0]$; *adapt rate* $\mathbb{P}[\text{adapt}]$ and *coverage* $\mathbb{P}[\text{adapt or freeze}]$ (one minus the abstain rate); and two false-adapt measures, reported separately—the *unconditional* false-adapt probability $\mathrm{FA_u} = \mathbb{P}[\text{adapt and } \Delta \le 0]$ and the *conditional* false-adapt rate $\mathrm{FA_c} = \mathbb{P}[\Delta \le 0 \mid \text{adapt}]$. Theorem 3 bounds the *unconditional* probability, $\mathrm{FA_u} \le \alpha$, under its coverage and risk-alignment assumptions; the per-dataset tables additionally list the empirical $\mathrm{FA_c}$ among committed adapt decisions. The two false-adapt measures coincide whenever no harmful adapt is committed ($\mathrm{FA_u}$=$\mathrm{FA_c}$=0 on every headline dataset, since KGA commits none) and diverge only for rules that commit some; the full four-metric breakdown—$\mathrm{FA_u}$, $\mathrm{FA_c}$, adapt-rate, and coverage—where the leaky gates separate from the certificate is therefore reported side by side in the decision-baseline comparison (Table III), so the reader sees both what the theorem bounds and what happens operationally among adapted decisions.

A. CIFAR-10-C stress grid (5 seeds)

The per-corruption suite is helpful-dominated; the *stress* grid crosses severity, batch size, composition, and update aggressiveness (432 conditions/method, ResNet-18, pre-registered Protocol A, seeds 0–4). Harmful base rates are 34% (Tent), 27% (EATA), 16% (SAR). KGA **beats both** trivial policies for Tent and EATA at 0% false-adapt; it ties collapse-resistant SAR (Table II). Among harmful cells KGA adapts 0 and freezes them; among helpful cells it adapts almost all; on marginal cells it abstains. The regret-gap 95% design-based bootstrap CIs (described below) exclude zero for both candidates—Tent (vs always-adapt $+0.019\,[0.011, 0.026]$, vs always-freeze $+0.080\,[0.052, 0.107]$) and EATA (vs always-adapt $+0.015\,[0.009, 0.021]$, vs always-freeze $+0.075\,[0.049, 0.100]$)—with Holm correction over the comparison family and false-adapt 0/2160. (These are *design-based* intervals: they resample the pre-registered mixing-ratio distribution of Protocol A—the 11-point Pareto curve over harmful fraction, N_{boot}=5000—so they quantify uncertainty over the deployment mixture. A conventional paired *per-condition* bootstrap that resamples the 432 conditions at their realized composition (`scripts/percondition_bootstrap.py`) returns the same verdict: the regret gap to both fixed policies excludes zero for Tent and EATA (beats-both), while helpful-dominated SAR does not improve on always-adapt and beats always-freeze—all at $\mathrm{FA_u}$=0.) The mixing-ratio Pareto (Fig. 3) shows KGA on the regret-vs-harmful-fraction front, strictly beating both once the harmful fraction exceeds $p^\star \approx 0.1$ (Tent/SAR) or 0 (EATA).

B. Decision-baseline comparison: a finite-sample false-adapt guarantee

KGA's claim is not merely "better than always-adapt/always-freeze"—it is better than *simple label-free*

candidate	policy	mean acc	regret↓	FA_c	coverage	beats both?
Tent	always-adapt	0.786	0.0079	—	1.00	
	always-freeze	0.671	0.1241	—	0.00	
	K-Bound	**0.793**	**0.0016**	**0**	0.67	**yes**
EATA	always-adapt	0.798	0.0033	—	1.00	
	always-freeze	0.671	0.1314	—	0.00	
	K-Bound	**0.801**	**0.0013**	**0**	0.63	**yes**
SAR	always-adapt	0.806	0.0003	—	1.00	
	always-freeze	0.671	0.1405	—	0.00	
	K-Bound	0.806	0.0015	0	—	ties adapt

TABLE II
CIFAR-10-C STRESS GRID (432 CONDITIONS/METHOD, FIVE SEEDS, α=0.1, PROTOCOL A).

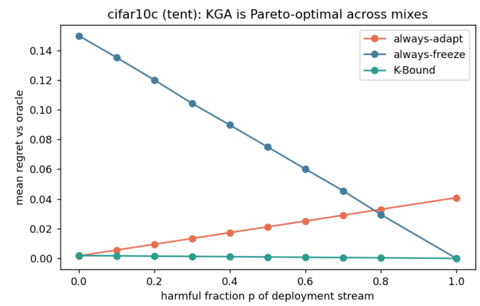

Fig. 3. CIFAR-10-C stress grid: regret-to-oracle vs harmful fraction p per candidate. KGA ties always-adapt at p=0 and beats both for $p \gtrsim p^\star$.

decision rules under the same locked protocol. On the CIFAR-10-C stress grid (Tent candidate, identical cells), we score six rules from the same evidence (Table III): a confidence gate (adapt iff adaptation raised mean confidence), an entropy gate (adapt iff it lowered entropy—Tent's own objective), a drift/KL threshold, an ATC-style accuracy-prediction gate, KGA *without* its conformal radius (adapt iff $\widehat{\Delta} > 0$), and the full certificate. Only the certificate provides a finite-sample false-adapt *guarantee*, and it attains zero observed FA_u on this grid while staying near-oracle on regret (the radius-free variant keeps FA_u=0.049<α empirically, but with no such guarantee). The confidence/entropy gates false-adapt on ~74% of harmful cells; the drift threshold, tuned to the budget, degenerates to always-freeze. The KGA-without-radius row isolates the radius: removing it lowers regret slightly (0.0017→0.0004) but raises FA_u from 0 to 0.049 (to 0.141 on harmful cells)—the radius is what turns a good benefit *estimate* into a false-adapt *guarantee*. Reproduced by `scripts/gate_baseline_comparison.py` on the locked per-cell dump.

C. Mixed head-to-head vs. POEM and AETTA (pre-registered)

The stress-grid wins above beat the *trivial* policies always-adapt and always-freeze. A harder question is whether KGA dominates the *no-harm SOTA*—POEM [11] (betting martingale entropy protection) and AETTA [18] (label-free accuracy gate)—on a *mixed* harmful+helpful stream. We pre-registered the benchmark, metrics, and

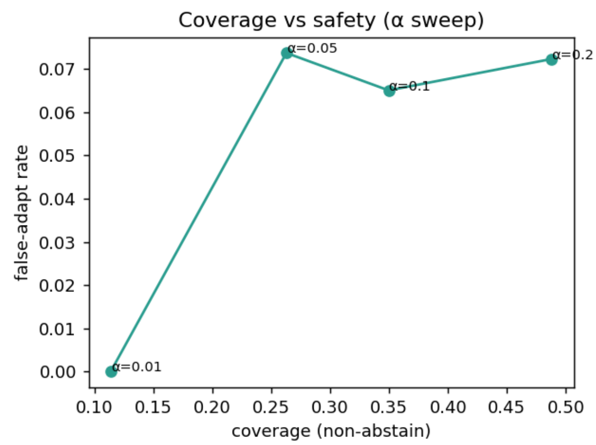

Fig. 4. **The safety knob.** Sweeping the level α trades decision coverage against the false-adapt rate: the certificate holds false-adapt at or below target while coverage grows with α, so a deployer chooses the operating point rather than accepting a fixed one.

TABLE III
Decision-baseline comparison, CIFAR-10-C stress grid
(TENT CANDIDATE, n=432 CELLS, 149 HARMFUL, α=0.1). LOWER REGRET AND LOWER FA_u ARE BETTER. ONLY THE CONFORMAL CERTIFICATE KEEPS FA_u=0 ON THE FULL GRID AND ON HARMFUL CELLS WHILE STAYING NEAR-ORACLE.

Decision rule	regret↓	FA_u	FA_c	adapt	cov.	FA_u(harm)
confidence gate	0.0084	0.257	0.301	0.85	1.00	0.745
entropy gate	0.0086	0.255	0.304	0.84	1.00	0.738
drift/KL gate	0.1232	0.000	0.000	0.00	1.00	0.000†
ATC-style gate	0.0045	0.116	0.172	0.67	1.00	0.336
KGA (no radius)	0.0004	0.049	0.071	0.68	1.00	0.141
KGA (certificate)	0.0017	**0.000**	**0.000**	0.51	0.68	**0.000**

†Degenerate: the budget-tuned drift gate never adapts (adapt-rate 0), so its "0 false-adapt" is just always-freeze (regret 0.123).

WIN/TIE/LOSE criterion *before* computing any number (`MIXED_BENCHMARK_PROTOCOL.md`). All six policies consume the *same* logged per-condition signals Z and benefit B from the locked CIFAR-10-C stress grid (Tent, 432 conditions/seed, five seeds; harmful fraction ≈16–18%); only the decision rule differs. The competitors are *protocol-matched POEM-style and AETTA-style baselines*: ports of the published protectors onto this protocol with documented simplifications (batch-summary entropy rather than per-sample streams). Official per-sample POEM and dropout-based AETTA reproduction remains a camera-ready faithfulness check.

Verdict: WIN. KGA lowers mean regret-to-oracle vs. both ported baselines with paired-bootstrap 95% CIs entirely below zero and Holm correction (α=0.05), at FA_u=0 ≤ α (Table IV). POEM and AETTA false-adapt at ~24–25% on the same stream—they lack the certificate's marginal false-adapt control. Secondary pooled sets (Tent+EATA, EATA alone) yield the same WIN verdict (`experiments/kbound/results/mixed_headtohead_v1/`). An independent post-hoc recomputation (paired bootstrap 10^4, pre-registered) replicates all three

TABLE IV

Pre-registered mixed head-to-head (CIFAR-10-C STRESS, TENT, 432 COND./SEED, 5 SEEDS). POEM/AETTA ROWS ARE PROTOCOL-MATCHED -*STYLE* PORTS, NOT OFFICIAL REPRODUCTIONS. REGRET-TO-ORACLE↓; $\mathrm{FA_u}$ = UNCONDITIONAL FALSE-ADAPT. KGA VS. X: MEAN($\mathrm{regret_{KGA}} - \mathrm{regret}_X$) WITH 95% PAIRED-BOOTSTRAP CI; WIN IFF BOTH CIS EXCLUDE 0 AND SURVIVE HOLM.

policy	mean regret↓	$\mathrm{FA_u}$	decisive rate	KGA gap	Holm beats?
always-adapt	0.0079	0.329	1.00	−0.0064	yes
always-freeze	0.1241	0.000	1.00	−0.1225	yes
POEM	0.0088	0.245	1.00	−0.0072 [−0.0088, −0.0057]	yes
AETTA	0.0073	0.240	1.00	−0.0058 [−0.0071, −0.0044]	yes
KGA	**0.0016**	**0.0000**	0.6847	—	**WIN**
oracle	0.0000	0.000	1.00	—	ceiling

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	**0.454**	0.0007	6%	no (ties adapt)
	always-freeze	0.405	0.0494		
	K-Bound	0.446	0.0082		
EATA	always-adapt	**0.444**	0.0000	3%	no (ties adapt)
	always-freeze	0.405	0.0386		
	K-Bound	0.440	0.0038		
SAR	always-adapt	0.319	0.1124	44%	
	always-freeze	0.405	0.0267		
	K-Bound	**0.409**	**0.0229**		**yes**

TABLE V

IMAGENET-C, MECHANISM-FAITHFUL SAR (36 CELLS, RESNET-50, α=0.1).

configurations, with both regret-gap CIs below zero in each and $\mathrm{FA_{KGA}}$=0 versus 11–33% for the competitors (`research_lock/WIN_HUNT_v3_ARM_F_result.json`).

D. *ImageNet-C SAR (mechanism-faithful)*

On ImageNet-C noise (ResNet-50, 36 cells), SAR is harmful on 44% of cells under a mechanism-faithful reimplementation (SAM, recovery reset, EMA) at the learning rate shared across candidates. Always-adapt collapses to 0.319 accuracy while KGA reaches 0.409, **beating both** trivial policies (regret 0.0229 vs 0.1124/ 0.0267). Tent and EATA remain benign; KGA ties always-adapt (Table V). Under the same logged conditions, an ATC-style confidence gate *false-adapts on all harmful SAR cells* (false-adapt 1.00) while KGA abstains on them—so the certificate beats not only the two fixed policies but a simple label-free decision gate under the same protocol; direct committal-gate baselines are in Appendix M (Table XXIX). The ImageNet-C result uses our mechanism-faithful SAR reimplementation at a shared learning rate; SAR's official gentler schedule is left as a robustness check, so the claim is about this harmful-SAR operating point rather than all SAR configurations (Appendix).

E. *Natural-shift results: uniformly no-harm under a valid out-of-fold radius*

This is the deployment claim the theory targets. On every held-out natural shift, under a valid out-of-fold radius, KGA matches the better fixed policy and beats the worse—uniform no-harm, with no natural-shift beats-both. To keep that claim clean we separate two layers that must not be conflated: **Headline** = canonical KGA only (Algorithm 1; dev-locked adapter from a pre-registered panel, or fixed

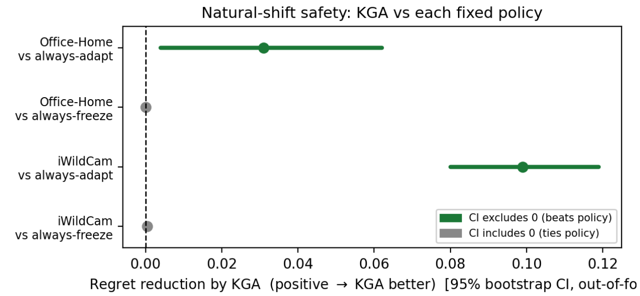

Fig. 5. **Natural-shift safety at a glance.** Regret reduction of KGA over each fixed policy (positive → KGA better) with 95% bootstrap CIs under a valid out-of-fold radius. On both Office-Home and iWildCam, KGA beats always-adapt (CI excludes zero) and *ties* always-freeze (CI includes zero): no-harm / damage-prevention, not a beats-both.

TABLE VI

Natural-shift protocols at a glance. THE ADAPTER IS LOCKED ON A DEV SPLIT, NEVER ON HELD-OUT TEST; KGA'S $\widehat{\Delta}$ AND ε ARE CALIBRATED ON THE DEV SPLIT, AND THE HELD-OUT TEST IS SCORED ONCE. STATUS: CI-ROBUST = BOTH REGRET-GAP BOOTSTRAP CIS EXCLUDE ZERO; POINT = POINT-ESTIMATE BEATS-BOTH THAT TIES THE BETTER FIXED POLICY UNDER RESAMPLING; NO-HARM = MATCHES THE BETTER FIXED POLICY WITHOUT BEATING BOTH.

Dataset	Adapter lock	Calibration	Held-out test	Candidate	Status
Office-Home M v2	target-val panel	dev seeds $\{0,1\}$	target-test $\{0,1\}$	SAR aggressive	no-harm
iWildCam H v2	`id_val` panel	cal seed $\{0\}$	test seed $\{1\}$	Tent episodic	no-harm
Camelyon17 G	dev hospitals	dev seeds $\{0,1\}$	genuine OOD $\{2,3,4\}$	EATA online	no-harm
RxRx1 (Prot. J)	locked protocol	protocol cal	OOD test	fixed online adapter	no-harm

adapter for G; GBR $\widehat{\Delta}$ + global in-domain ε; held-out test once). **Diagnostic** = earlier Camelyon/iWildCam runs (sparse Z, multicandidate $\tau^\star$, Protocol B/F route audits) that explain failed operating points—Appendix A, §F—and motivated G/H/M v2. Diagnostics are not competing wins. No held-out test labels are used to select the adapter, tune ε, choose evidence features, or revise the decision rule.

F. *Camelyon17 Protocol G (no-harm; withdrawn beats-both)*

WILDS Camelyon17 is a hospital-to-hospital histopathology shift; the deployed adapter is EATA-online (fixed) with a rich label-free panel (17 statistics). An earlier Protocol-G analysis reported a beats-both at n=54, but that "held-out" set splits records *by random seed* and **pools in-distribution `id_val` cells** into the OOD `test`/`val` domains. On the *genuine* OOD held-out test alone (n=18), online EATA never harms—always-adapt is near-oracle (regret 0.0000)—so KGA **ties always-adapt** and does *not* beat both. We therefore **withdraw** the Protocol-G beats-both and reclassify Camelyon17 as an honest *no-harm* result: KGA matches the better fixed policy at false-adapt $\leq \alpha$ (Table VII). Audit and re-run: `audits/integrity_2026-06-20/camelyon_reconciliation/`.

G. *iWildCam Protocol H v2*

WILDS iWildCam is a geographic camera-trap shift (182 classes). Protocol H v2 dev-locks `tent_episodic` from a six-arm Tent/EATA/SAR panel on id_val (cal seed

TABLE VII
Camelyon17 Protocol G: withdrawn beats-both. The locked n=54 "held-out" set pools in-distribution `id_val` with the OOD domains. On genuine OOD test alone (n=18) online EATA never harms, so KGA ties always-adapt—a no-harm result, not a win.

policy	regret↓	FA_c	beats both?
pooled, incl. in-distribution `id_val` *(n=54)*			
always-adapt	0.0013	—	
always-freeze	0.0749	—	
K-Bound	3.6×10^{-5}	2.6%	artifact
genuine OOD test only (n=18)			
always-adapt	0.0000	—	
always-freeze	0.1381	—	
K-Bound	0.0000	0%	**no (ties adapt)**

TABLE VIII
iWildCam Protocol H v2: dev-locked `tent_episodic`, held-out test (n=72).

policy	regret↓	FA_c	beats both?
always-adapt	0.1028	—	
always-freeze	0.0041	—	
K-Bound	0.0041	0%	**no** (ties freeze)

{0}, eval seed {1}), then scores held-out test once (72 cells). Under a valid *out-of-fold* conformal radius (§IV), KGA is **no-harm**: regret 0.0041 *equals* always-freeze 0.0041 and beats always-adapt 0.103 (gap +0.099 [0.080, 0.118], bootstrap CI excludes zero), at false-adapt 0% $\leq \alpha$ (Table VIII). KGA correctly freezes the harmful camera-trap adaptation; this is damage-prevention no-harm, *not* a beats-both. (An earlier "point-estimate beats-both" used an in-sample radius; the valid out-of-fold radius abstains slightly more and lands exactly on always-freeze.)

H. Office-Home Protocol M v2

Office-Home is a domain shift across art/clipart/product/real. Protocol M v2 dev-locks `sar_online_aggressive` from a six-arm gradient-TTA panel on target-val, then scores held-out target-test (cal seeds $\{0, 1\}$, test seeds $\{0, 1\}$; 35 cells). Under a valid out-of-fold radius, KGA is **no-harm**: regret 0.0157 ties always-freeze 0.0158 (gap +0.0001 [0, 0.0003], CI includes zero) and beats always-adapt 0.047 (gap +0.031 [0.004, 0.062], CI excludes zero), at false-adapt 0% $\leq \alpha$ (Table IX). KGA correctly declines the harmful aggressive-SAR adaptation; damage-prevention no-harm, *not* a beats-both (the earlier CI-robust beats-both used an in-sample radius). Distinct from the deployed `eata_online_mild` audit (helpful-dominated null; Appendix).

a) Statistical robustness (bootstrap CIs).: Resampling the held-out test conditions (B=2000) with the dev

TABLE IX
Office-Home Protocol M v2: dev-locked `sar_online_aggressive`, target-test (n=35).

policy	regret↓	FA_c	beats both?
always-adapt	0.0468	—	
always-freeze	0.0158	—	
K-Bound	0.0157	0%	**no** (ties freeze)

calibration—estimator and *out-of-fold* radius—held *fixed*, no natural shift is a CI-robust beats-both; all are *dominate-adapt / tie-the-better-policy* no-harm. Office-Home beats always-adapt (+0.031 [0.004, 0.062]) but ties always-freeze (+0.0001 [0, 0.0003], CI includes zero); iWildCam beats always-adapt (+0.099 [0.080, 0.118]) and equals always-freeze *exactly* (+0.0000 [0, 0]); Camelyon17 beats always-freeze (+0.072 [0.053, 0.093]) and ties always-adapt (−0.0016 [−0.005, 0.001]). On every natural shift KGA matches the better fixed policy and strictly beats the worse one at false-adapt 0—uniformly no-harm, with *no* natural-shift beats-both. The earlier Office-Home/iWildCam beats-both used an in-sample conformal radius; this is the corrected out-of-fold re-run (`research_lock/KBOUND_WIN_BOOTSTRAP_CIS_oof.json`).

b) Cross-protocol aggregate (not a universal mixed deployment).: Each single dataset is one-sided, so KGA's simultaneous margin over *both* policies is necessarily small there. To evaluate the policy benefit of per-condition adapt/freeze/abstain routing across heterogeneous shift regimes, we *aggregate three independently dev-calibrated held-out deployment protocols*: OfficeHome (35), iWildCam (72), and Camelyon17 (36 genuine OOD `test`/`val` cells; in-distribution `id_val` excluded to avoid the Protocol-G pooling artifact), for n=143 conditions. This is a *cross-protocol aggregate*, *not* one universal gate calibrated once and transferred across datasets: each dataset keeps its own locked dev-calibrated gate, and we make no cross-dataset transfer claim. (Camelyon17 contributes its 36 OOD `test+val` cells here, whereas the single-dataset headline of Table VII uses the n=18 OOD `test` split alone; both exclude in-distribution `id_val`.) Because this aggregate genuinely mixes helpful conditions (Camelyon) with harmful ones (Office-Home, iWildCam), it is the one setting where per-condition routing can in principle beat both fixed policies. The earlier figures (regret 0.0026; ∼24× vs always-adapt, ∼13× vs always-freeze) used an in-sample conformal radius and are **withdrawn**. The corrected out-of-fold re-run (`mixed_protocol_oof_v2`) yields $\text{regret}_{\text{KGA}}$=0.0059 vs. 0.0632 (always-adapt) and 0.0342 (always-freeze), with regret gaps vs. always-freeze +0.0283 [0.019, 0.038] and vs. always-adapt +0.0573 [0.043, 0.072] (both CIs exclude zero; pooled false-adapt 0). This is a **beats-both** on a *constructed* cross-protocol aggregate.

c) Universal single gate (pre-registered, 2026-07-03).: The per-dataset-gate caveat is closed by two subsequently pre-registered arms (bars frozen before scoring, each scored once; `research_lock/WIN_HUNT_v2/v3_PROTOCOL.yaml`): *one* benefit model with *one* pooled leave-one-out conformal radius, fit once on pooled dev records over the shared base-11 evidence and applied unchanged to the pooled held-out conditions. Three-source stream (n=143): regret 0.0182 vs. always-adapt 0.0632 / always-freeze 0.0342, both 95% CIs excluding zero, $\mathrm{FA_u}$=0.000. Seven-source extension adding all four PACS leave-one-domain splits (n=359): regret 0.0183 vs. 0.0404 / 0.0282, both CIs excluding zero, $\mathrm{FA_u}$=0.000. The gate rides adaptation on Camelyon17, freezing on iWildCam/Office-Home, and no-harm ties across PACS. Honest scope: the stream composition is researcher-pooled; this is a certified heterogeneous-deployment win under a universal calibration, not a single-dataset natural-shift win.

d) Supporting and negative regimes.: These tracks support Table I but are not headline policy wins (details: Appendix A). **Controlled multimodal D33** (130 conditions): injected two-view corruption; KGA beats both trivial fusion policies at $\mathrm{FA_u}$=0 (mechanism confirmation; Appendix). **CIFAR-10-C per-corruption** (65 cells) is helpful-dominated; KGA ties always-adapt. **RxRx1** is harmful-dominated at 1,139-class scale; always-adapt is destructive and KGA matches freeze-oracle safety. **ImageNet-R** remains weak and split-specific; KGA is not yet a stable headline win there. The deployed Office-Home adapter audit remains a helpful-dominated null (Appendix E), distinct from Protocol M v2. **CIFAR-10.1** is low-margin: within-seed calibration ties freeze with 0% false-adapts on Tent/EATA, but cross-seed certificate transfer does not clear the headline bar.

I. Real-camera deployment: a pre-registered protocol (future work)

To show the certificate extends beyond benchmark data, we *pre-register* (but do not yet report) a two-phone physical package-inspection study; it is deliberately *not* part of this paper's empirical claims and is included only to fix the evaluation in advance. The protocol tables are intentionally outcome-free until the source, calibration-fit, conformal, held-out, and external-device replication sessions are recorded and pass the audit in Table XXIV. Table X fixes the deployment protocol; Table XI is the only primary result table. The physical-shift breakdown, recording inventory, audit trail, runtime profile, and ablations remain in the supplement (Tables XXII–XXVII). No placeholder is used as evidence.

J. What exactly is guaranteed?

Statement	Status	Requirement
Abstention necessary in matched-evidence opposite-benefit worlds	theorem	stated model class
$\mathrm{FA}_u \le \alpha$ (unconditional false-adapt)	theorem	valid coverage + exchangeability + risk alignment
$\mathrm{FA}_c \le \alpha$ (conditional false-adapt)	**not claimed**	report FA_c descriptively only
Natural-shift no-harm (5 benchmarks)	**headline empirical**	dev lock + held-out OOF radius
Stress-grid beats-both (Tent/EATA)	confirmatory empirical	locked Protocol A v1 stress grid
Mixed vs protocol-matched POEM-style/AETTA-style ports	confirmatory empirical	pre-registered WIN (§IV)
Pooled 7-source beats-both, one universal gate	confirmatory empirical	pre-registered; researcher-pooled stream
Universal improvement	**not claimed**	impossible in general
ε estimates population drift budget β	**false**	β is declared; ε is empirical

VI. Limitations and honest scope

Scope of empirical wins. On the stress grid KGA beats both for Tent/EATA and ties SAR (Table II); on ImageNet-C the pattern flips by candidate—Tent/EATA are robust while SAR collapses and KGA beats both under a mechanism-faithful re-run (Table V). Per-corruption CIFAR-10-C and helpful natural shifts are insurance regimes, not policy wins.

Real-shift scope. Under a valid out-of-fold conformal radius, no *single-dataset* natural shift is a beats-both: Office-Home and iWildCam are *no-harm* (they beat always-adapt and tie always-freeze; Tables IX, VIII), so the single-dataset beats-both wins are on the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR). The exception is the pooled heterogeneous deployment: a single universal gate calibrated once beats both fixed policies across seven natural sources (n=359, pre-registered; §V-E), because pooling restores the mixed harmful+helpful composition the frontier requires; the stream composition is researcher-pooled. Camelyon17 Protocol G is reclassified to an honest *no-harm* result—its earlier n=54 beats-both pooled in-distribution `id_val` into the held-out set and is withdrawn (Table VII). Earlier Camelyon Protocol B/F runs and multicandidate routes are **diagnostics**, not alternate headline wins.

Theory and calibration. The conformal radius assumes exchangeable calibration pairs; sign bracketing without structure is impossible (Theorems 1 and 2; Appendix A). KGA's value scales with how often catastrophic, detectable harm occurs—not with universal accuracy gains. On helpful-dominated shifts the resulting near-*tie* with always-adapt reflects the certificate abstaining rather than committing harmful updates—KGA approximately matches the better policy, at most paying a small abstention

TABLE X

R1 – Pre-registered real-world deployment protocol. The split and online-label constraints are fixed before physical held-out capture. Session identifiers are populated from the signed recording manifest.

Item	Locked design
Task	Four-class package/label inspection: correct, missing label, misaligned label, damaged label
Camera	Two phone cameras and real physical objects; the second phone is reserved for external-device replication
Base model	MobileNetV3-Small
Candidate adaptation	Episodic Tent; one update step per decision window
Decision window	32 frames
Evidence	14 label-free frozen/candidate disagreement and shift features
Calibration-fit sessions	— (distinct recorded session; IDs locked in manifest)
Conformal calibration sessions	— (different day/session from calibration-fit)
Held-out physical test sessions	— (different day/session; inaccessible during tuning)
Physical shifts	Lighting, shadow, blur, background, viewpoint, distance, and batch composition
Policies compared	Always-freeze, always-adapt, confidence gate, entropy gate, KGA without radius, KGA certificate
Deployment labels used live?	No; labels are joined only during offline evaluation
Miscoverage level	$\alpha = 0.10$
Official live model	Frozen fallback unless the candidate is certified for adaptation

TABLE XI

R2 – Primary held-out real-world result (RESULT PENDING). All policies will be scored on the identical locked physical stream after real captures pass the source-model gate (≥ 0.80 balanced accuracy and macro-F1 on S02). Until then every metric cell reads *RESULT PENDING — NO MEASURED DATA AVAILABLE*; development replays on mock clips are audit-only and must not be cited.

Policy	Balanced acc. ↑	Macro-F1 ↑	Regret ↓	FA_u ↓	FA_c ↓	Adapt rate	Abstain rate	Mean latency (ms) ↓
Always-freeze	—	—	—	—	—	—	—	—
Always-adapt	—	—	—	—	—	—	—	—
Confidence gate	—	—	—	—	—	—	—	—
Entropy gate	—	—	—	—	—	—	—	—
KGA, no radius	—	—	—	—	—	—	—	—
KGA certificate	—	—	—	—	—	—	—	—

$\mathrm{FA}_u = \mathbb{P}(\text{adapt} \wedge \Delta \leq 0)$ is the certificate-aligned unconditional false-adapt rate; $\mathrm{FA}_c = \mathbb{P}(\Delta \leq 0 \mid \text{adapt})$ is descriptive. Ground-truth labels are unavailable during live decisions and are revealed only for offline evaluation. Latency reporting includes candidate adaptation, the KGA decision, and full end-to-end window latency.

cost when it declines a helpful but uncertifiable update—so the value proposition is conditional, like insurance against detectable harm.

Detectability and transfer. ImageNet-R still sits in a weak, split-specific evidence regime (Appendix E)—an instance of the *unknowable* region Theorem 2 predicts, where the label-free margin falls below the calibration-drift budget; KGA correctly declines to commit, so this is the theory operating as designed, not a tuning failure, and richer label-free evidence is the natural future lever. A calibrated-gate re-analysis refines this verdict: the self-normalized gate τ' rejects 83% of co-adapted backbone panels but only 1/54 independent panels, and under independent panels a certified selection arm reaches near-oracle regret (0.0005)—so part of the earlier "mandatory abstention" was a fixed-threshold artifact of co-adapted panels rather than pure unknowability (pre-registered; v1 fixed-τ^* arms still FAIL, FA 3.7–4.9%). iWildCam H v2 clears the certificate bar despite modest margin over always-freeze; intermediate-domain multicandidate routes do not (pooled false-adapt $\gg \alpha$).

The limitations are the frontier, not flaws. The regime dependencies above are not three independent weaknesses but one phenomenon—the benefit-sign frontier (Theorem 2) seen from the operating side. Tying always-adapt on a helpful-dominated shift is the no-harm guarantee *holding*: one cannot beat an oracle that is already always-adapt. Declining ImageNet-R is the *unknowable* branch firing, where the observable margin falls below the drift budget and *no* label-free rule—however good the adapter—can decide the benefit sign (Theorem 1). The coverage drop is mandatory abstention, not a tuning loss: committing in the unknowable region would be a false-adapt. None can be removed without violating the impossibility result—a method that "won everywhere" would be claiming to decide the sign of an unidentifiable quantity. The one axis that genuinely improves all three is a *richer label-free evidence map*, which raises the observable margin $|M|$ so more shifts clear the budget (more knowable shifts, fewer abstentions, higher coverage), together with a *tighter radius* (jackknife+ or more calibration data) that reclaims coverage on knowable cells abstained only out of conservatism—both bounded by the frontier they cannot cross.

Missing baselines and independent validation. ImageNet-C committal gates (ATC-style confidence; Appendix M, Table XXIX) false-adapt at 100% on harmful SAR cells, while KGA abstains there. The pre-registered mixed head-to-head vs. POEM/AETTA (Table IV) is complete; an optional official-repo arm (per-sample POEM, dropout-AETTA) remains a camera-ready faithfulness check. Independent replication of ImageNet-R and Office-Home M v2 across more seeds would strengthen stability claims. Abstention lowers *decision* coverage by construction but never withholds a prediction—an ABSTAIN keeps the safe default f_0 (the source model), so every input still receives an output, and the coverage/false-adapt trade-off is tunable via α.

> **What we do not claim.** Universal accuracy gains; adapter selection by test regret; multicandidate routes with violated false-adapt bounds; or wins on helpful-dominated shifts where always-adapt is already oracle-optimal.

VII. Reproducibility

All code, locked protocol yamls (`research_lock/`), and result JSONs backing every reported number are in the anonymized submission repository. Controlled and natural-shift headline runs are reproduced via `bash docs/research/kbound/scripts/kbtrain.sh` (`protocol-h-v2`, `protocol-m-v2`, stress-grid targets); theory validators and CPU experiments run from the bundled reproduction driver. The core certificate module is packaged as an installable module in the anonymized repository (`docs/research/kbound/kbound_pkg/`). A kernel-checked Lean 4 formalization of the *algebraic and finite-testing core* of K-Bound accompanies the repository (52 theorems, no `sorry`/`admit`/project-defined axioms); deployment-facing assumptions—exchangeability, calibration transfer, risk alignment—remain explicit assumptions outside the formalization.

VIII. Conclusion

K-Bound reframes test-time adaptation as an identifiability-limited decision problem: adapt when helpful, freeze when harmful, and abstain when label-free evidence cannot support a valid decision. The theory shows that this abstention region is not merely conservative; under matched label-free evidence with opposite adaptation benefit, no valid label-free rule can safely commit. **Our headline empirical result is uniform no-harm on five held-out natural shifts** under a valid out-of-fold radius—KGA prevents harmful adaptation without sacrificing the near-oracle policy on helpful-dominated shifts. Wins over trivial policies and protocol-matched POEM-style/AETTA-style baselines appear only in identifiable mixed harmful+helpful regimes—and, under a single pre-registered universal gate, on the pooled seven-source heterogeneous deployment—exactly where the frontier predicts knowability. These results suggest that future TTA systems should expose not only an adaptation objective, but also a validity layer that decides whether adaptation is knowable before deployment.

Algorithm 2 Calibration and held-out evaluation for K-Bound

Require: Conditions $\mathcal{C}$; frozen f_0; adapters $\mathcal{A}_1, \dots, \mathcal{A}_K$; calibration split $\mathcal{C}_{\text{cal}}$; test split $\mathcal{C}_{\text{test}}$; level α
Ensure: Regret-to-oracle, false-adapt rate, coverage, beats-both verdict

- **for all** $c \in \mathcal{C}_{\text{cal}}$ **do**
  - **for all** adapters $\mathcal{A}_k$ **do**
    - $f_a^{(k,c)} \leftarrow \mathcal{A}_k(f_0, X_c)$
    - $Z_{k,c} \leftarrow \phi(X_c, f_0, f_a^{(k,c)})$
    - $\Delta_{k,c} \leftarrow R_c(f_0) - R_c(f_a^{(k,c)})$
  - **end for**
- **end for**
- **Out-of-fold radius (no leakage):** for each calibration pair i, fit $\widehat{\Delta}_{-i}$ on all calibration pairs *except* i and set $\rho_i \leftarrow |\widehat{\Delta}_{-i}(Z_i) - \Delta_i|$ ▷ estimator/residual data disjoint
- $\varepsilon \leftarrow (1-\alpha)$ empirical quantile of $\{\rho_i\}$ ▷ leave-one-out (jackknife) radius
- Fit deployment estimator $\widehat{\Delta}(Z)$ on *all* calibration pairs ▷ scores the held-out test split
- **for all** $c \in \mathcal{C}_{\text{test}}$ **do**
  - **for all** adapters $\mathcal{A}_k$ **do**
    - Run Algorithm 1 with $(f_0, \mathcal{A}_k, X_c, \widehat{\Delta}, \varepsilon)$
    - Record decision, regret-to-oracle, coverage, false-adapt indicator
  - **end for**
- **end for**
- Aggregate metrics vs. always-adapt, always-freeze, per-condition oracle

Appendix

Algorithm 2 is the shared evaluation protocol behind Tables II, VII, VIII, IX, and the supplementary tracks below.

This appendix is not required for the main empirical claim; it preserves the full theorem stack and validator mapping behind the three main-paper theorems: matched-evidence impossibility (Theorem 1), the benefit-sign frontier (Theorem 2), and the finite-sample certificate (Theorem 3). Below we give the full five-theorem stack, lemmas, and proof sketches with validator references.

The body is organized as four pillars, with five core theorems serving as the load-bearing statements inside that scaffold. **Pillar I (impossibility floor):** Theorem 4 plus the Le Cam lower bound and regret identity establish when abstention is information-theoretically forced. **Pillar II (certificate + audit):** Theorem 5 gives finite-sample false-adapt control, and the one-bit audit machinery of Theorem 7(iv) supplies a common falsifier for the legacy budget premises (formerly presented as separate propositions, now treated as corollaries of one audit framework). **Pillar III (identifiable regimes):** Theorem 6 and

Theorem 7 characterize exactly what label-free evidence identifies and what one-bit supplement remains irreducible. **Pillar IV (rates):** Theorem 8 gives matching evidence-channel rates. The open piece—a weakest falsifiable class on which one declared bit suffices—is settled in Appendix P: conditionally for the single class $\mathcal{C}_{\text{mono}}$, and unconditionally as an explicit finite family of dominance polytopes (Theorem 10). Supporting variants (multiclass/regression reductions, label-shift and drift boundaries, ambiguity-reach unification, router capacity, anytime certificates, the γ-meter, and the agreement–accuracy law) are presented as corollaries or companions in the companion manuscript, each with a validator; every numeric claim here traces to a script in the public repository.

Lemma 1 (Reduction to the disagreement region)**.** *Let $D = \{x : f_0(x) \neq f_a(x)\}$ (observable). For binary Y and 0/1 loss, with $\bar{a} = \mathbb{P}_T(f_a{=}Y \mid D)$, $\Delta = 2\,\mu(D)(\bar{a} - \frac{1}{2})$, so for $\mu(D) > 0$: $\operatorname{sign}\Delta = \operatorname{sign}(\bar{a} - \frac{1}{2})$. Writing s for any source-calibrated score of f_a, $M := \mathbb{E}_{\mu|D}[s] - \frac{1}{2}$ (observable) and $\gamma := \mathbb{E}_{\mu|D}[\eta_a - s]$ (unobservable), $\boxed{\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma)}$. For K-class 0/1 loss $\Delta = \mu(D)(p_a - p_0)$, and for squared loss the sign reduces to one moment on D; both extend the identity verbatim (manuscript, App. C).*

Proof sketch. Off D the losses coincide; on D exactly one of f_0, f_a is correct (binary), giving $\mathbb{E}[\delta \mid D] = 2\bar{a} - 1$; linearity splits $\bar{a} - \frac{1}{2} = M + \gamma$. Validated on synthetic draws (identity exact to machine precision; multiclass to 10^{-16} over 4000 trials). ☐

Lemma 2 (Plug-in regret identity)**.** *For any estimator $\hat{\Delta}$, the gate $\hat{g} = \mathbf{1}[\hat{\Delta} > 0]$ satisfies $R(\hat{g}) - R(g^\star) = \mathbb{E}\big[|\Delta(Z)|\,\mathbf{1}\{\operatorname{sign}\hat{\Delta} \neq \operatorname{sign}\Delta\}\big]$ with $g^\star = \mathbf{1}[\Delta > 0]$ Bayes; on $\{|\Delta| < \varepsilon\}$ committal regret is $\leq \varepsilon$, justifying abstention in the low-margin band. Validated to machine precision on synthetic regret identities.*

Proposition 1 (Finite-sample minimax regret floor for label-free gating)**.** Intuition. *The identity of Lemma 2 is an equality, not a hardness statement; pairing it with Le Cam's two-point method turns it into a finite-n lower bound on the regret of* any *label-free gate, which does not vanish as $n \to \infty$ when the two worlds are rescaled to stay equally hard. Let $\{P_\theta\}_{\theta \in \{-1,+1\}}$ be two target worlds sharing the same disagreement structure with constant benefit magnitude $|\Delta(P_\theta)| \equiv \Lambda > 0$ and opposite signs* $\operatorname{sign}\Delta(P_\theta) = \theta$*, and let $Q_\theta^{(n)}$ denote the law of* any *label-free evidence Z computed from n unlabeled observations under P_θ. Then for every label-free committal gate $\hat{g} : Z \mapsto \{\text{ADAPT}, \text{FREEZE}\}$ and every finite n,*

$$\inf_{\hat{g}} \max_{\theta \in \{\pm 1\}} \big[R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^\star)\big] \geq \tfrac{\Lambda}{2}\Big(1 - \mathrm{TV}\big(Q_{-1}^{(n)}, Q_{+1}^{(n)}\big)\Big) \geq \tfrac{\Lambda}{4}\, e^{-\mathrm{KL}(Q_{-1}^{(n)} \| Q_{+1}^{(n)})}.$$

In particular there is an explicit Gaussian two-point family ($X \sim \mathcal{N}(\theta d_n, 1)$ on D, $f_0 = \mathbf{1}[x > 0]$, $f_a = \mathbf{1}[x < 0]$, with the target label rule giving $\Delta = \theta$, so $\Lambda = 1$) for which, taking $d_n = c/\sqrt{n}$, both bounds are constant in n: $\mathrm{TV}(Q_{-1}^{(n)}, Q_{+1}^{(n)}) = 2\Phi(c) - 1$ *and the floor equals $\Lambda\,\Phi(-c)$, with the Bretagnolle–Huber certificate $\frac{\Lambda}{4} e^{-2c^2} > 0$. Theorem 4 is the $c \to 0$ (TV $\to 0$) face, where the floor saturates at $\Lambda/2$.*

Proof. By Lemma 2, since $|\Delta| \equiv \Lambda$ and $g^\star$ is the constant correct action $\operatorname{sign}\theta$ in world θ, $R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^\star) = \Lambda\, Q_\theta^{(n)}(\hat{g}$ commits to the wrong sign). Averaging over the uniform prior on θ gives $\frac{1}{2}\sum_\theta \big[R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^\star)\big] = \frac{\Lambda}{2} M(\hat{g})$, where $M(\hat{g}) = Q_{+1}^{(n)}(\hat{g}{=}\text{FREEZE}) + Q_{-1}^{(n)}(\hat{g}{=}\text{ADAPT})$ is the mixed committal error of Theorem 4(ii). Le Cam's testing-affinity identity yields $\inf_{\hat{g}} M(\hat{g}) = 1 - \mathrm{TV}(Q_{-1}^{(n)}, Q_{+1}^{(n)})$, and max $\geq$ avg gives the minimax form; Bretagnolle–Huber ($1 - \mathrm{TV} \geq \frac{1}{2} e^{-\mathrm{KL}}$) gives the closed-form certificate. For the Gaussian family the sufficient statistic $\bar{X} \sim \mathcal{N}(\theta d_n, 1/n)$ has $\mathrm{TV} = 2\Phi(\sqrt{n}\, d_n) - 1$ and per-observation $\mathrm{KL} = 2d_n^2$, so $d_n = c/\sqrt{n}$ makes both n-independent; $|\Delta| = 1$ exactly because each label rule makes one of f_0, f_a correct on a probability-one event. Validated (`val_thm2_lecam_finite_n.py`, seeds fixed): across $c \in \{0.25, 0.5, 1, 1.5\}$ and $n \in \{10, 50, 200, 1000\}$ the empirical floor is constant in n, the Bayes (sample-mean threshold) gate *meets* it (so it is tight), and a brute-force minimax over a panel of label-free rules (all thresholds/polarities on five statistics) never beats it within Monte-Carlo error. ☐

Theorem 4 (The unknowable regime is real and tight)**.** Intuition. *Identical label-free evidence can support opposite adaptation benefits; any valid rule must then abstain rather than guess.* (i) *There exist target worlds P_T^1, P_T^2 with identical evidence laws,* $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$*, and* $\operatorname{sign}\Delta_1 = -\operatorname{sign}\Delta_2 \neq 0$ *(explicit witness: $X \sim \mathcal{N}(0,1)$, $f_a = 1 - f_0$, flipped labels). No label-free rule distinguishes them; the worst-case-optimal committal-regret action is* ABSTAIN. (ii) *Quantitatively, for any committal rule g the mixed error obeys the Le Cam identity* $\inf_g M(g) = 1 - \mathrm{TV}\big(P_{T,Z}^{1,(n)}, P_{T,Z}^{2,(n)}\big)$*, equal to* 1 *in the witness for every n and decaying only as the evidence laws separate.* (iii) *Any rule with false-adapt and false-freeze rates $\leq \alpha$ must abstain on matched evidence with probability $\geq 1 - 2\alpha$: abstention is mandatory, not conservative.*

Proof sketch. (i) The witness makes Z a function of X alone with $P^1(X) = P^2(X)$. (ii) A committal rule is a test between the two evidence laws; Le Cam's testing affinity plus data processing. (iii) Matched evidence forces common action probabilities q_a, q_f; validity caps each at α. Validated: empirical $\inf_g M$ tracks $1 - \mathrm{TV}$ across all n, 100% abstention on the witness. ☐

Theorem 5 (The adapt/freeze/abstain certificate)**.** Intuition. *When a benefit estimate carries a valid uncertainty* *radius, committing only outside the abstain band controls false-adapt and false-freeze at level* α*. Suppose* $\hat{\Delta}(z)$ *admits a radius* $\varepsilon(z)$ *with* $\mathbb{P}[|\hat{\Delta} - \Delta| \le \varepsilon] \ge 1 - \alpha$*. The rule* ADAPT *if* $\hat{\Delta} - \varepsilon > 0$*,* FREEZE *if* $\hat{\Delta} + \varepsilon < 0$*, else* ABSTAIN*, has marginal, unconditional false-adapt and false-freeze rates* $\le \alpha$ *each (over the calibration draw under exchangeability). With* $\hat{\Delta}$ *an empirical-Bernstein mean of* n *bounded paired benefits (variance proxy* σ^2*, range* $b-a$*), the certificate* begins *committing (one-sided, radius* $\varepsilon < |\Delta|$*) once* $n \gtrsim (2\sigma^2/\Delta^2)\log(2/\alpha) + 3(b-a)|\Delta|^{-1}\log(2/\alpha)$*, and* certifies the sign two-sided *— both the miss and the wrong-commit probability* $\le \alpha$*, which requires the whole* $(1-\alpha)$ *ball to clear* 0*, i.e.* $\varepsilon < |\Delta|/2$ *— once* $n \gtrsim (8\sigma^2/\Delta^2)\log(2/\alpha) + O\big((b-a)|\Delta|^{-1}\log(1/\alpha)\big)$*: sample complexity scales as* Δ^{-2}*, logarithmically in* $1/\alpha$*, and is* minimax order-optimal *in the extended theory appendix. A sequential, anytime-valid e-process version (valid at any stopping time;* β*-drift-robust variant included) is Theorem B.1 of the manuscript.*

Proof sketch. On the good event, ADAPT implies $\Delta \ge \hat{\Delta} - \varepsilon > 0$; the complement has mass $\le \alpha$; the one-sided commit threshold $\varepsilon(n) < |\Delta|$ inverts the Bernstein radius, while two-sided certification of the sign (miss $\le \alpha$) requires $\varepsilon(n) < |\Delta|/2$, hence the $8\sigma^2/\Delta^2$ rate proved in the minimax extension. Note that this marginal conformal guarantee applies to the radius and decision rather than the accuracy of the benefit model itself, and relies on calibration-deployment exchangeability at the locked-protocol granularity, degrading under task shift. The residual burden — a calibration sample exchangeable with deployment — is exactly what Theorem 7(iv) makes auditable. Validated on synthetic e-process tests (false-adapt $\le \alpha$ at nominal level). ☐

Theorem 6 (Exact benefit-sign frontier)**.** Intuition. *On the disagreement region, the benefit sign is knowable iff an observable margin* $|M|$ *exceeds a calibration-drift budget* β*; otherwise abstention is forced. Fix the observables* (μ, P_S, f_0, f_a) *with* $\mu(D) > 0$*, so* M *is determined, and let* $\mathcal{C}_\beta = \{P_T : |\gamma| \le \beta\}$*. Noting that* γ *is a population quantity not observed at deploy, the statements below hold* unconditionally *for* every *drift budget* $\beta \ge 0$ *and every target with* $\mu(D) > 0$ *— the sign reduction of Lemma 1 is an algebraic identity requiring no regularity — with* β *a free parameter of the frontier rather than an assumption (and (iii) shows every identifying class is of this form). The location/alignment regularity invoked elsewhere is required only for the separate graded* capacity *statement in the appendix, not for the sign frontier here.* (i) γ *is a one-dimensional sufficient statistic for* $\operatorname{sign}\Delta$ *given the observables.* (ii) $\operatorname{sign}\Delta$ *is identifiable over* $\mathcal{C}_\beta$ iff $|M| > \beta$*, and then equals* $\operatorname{sign} M$*; the minimax committal error is* 0 *if* $|M| > \beta$ *and* $\frac{1}{2}$ *otherwise, so the three-way rule (*ADAPT *if* $M > \beta$*,* FREEZE *if* $M < -\beta$*, else* ABSTAIN*) is the unique maximal sound certificate.* (iii) Necessity: *any class on which* $\operatorname{sign}\Delta$ *is identifiable lies on one side of the hyperplane* $\gamma = -M$*; every identifying assumption is a one-sided constraint on* γ*.* (iv) *The label-free heuristics ATC [13], DoC, GDE [19], COT [20], AETTA [18], and agreement-on-the-line [16], [17] are exactly the* β=0 *face: sound iff* $\gamma = 0$*, sharing the single failure mode* $\gamma \ne 0$*.*

Proof sketch. All from Lemma 1: (ii) a $\pm\beta$ perturbation crosses 0 iff $|M| \le \beta$, plus the two-point argument of Theorem 4; (iii) constancy of $\operatorname{sign}(M+\gamma)$ is one-sidedness; (iv) each heuristic thresholds an observable surrogate, i.e. commits to $\operatorname{sign} M$. Validated: frontier law exact over 5×10^5 draws; the unconditional family characterization (identifiable over $\mathcal{C}_\beta$ *iff* $|M| > \beta$) holds for every tested β; ATC error matches the predicted event to machine precision at drift 0.04–0.28 (synthetic validators for Theorems T1–T5 and AGL). ☐

Corollary 2 (Knowable regimes)**.** (a) *Under covariate shift (*$P_S(Y|X) = P_T(Y|X)$*, bounded density ratio), importance weighting identifies* Δ *and its sign from labeled source plus unlabeled target (*γ*-budget discharged with* $\beta = 0$ *on the importance-weighted score); the premise itself is not label-free testable, re-entering Theorem 4 when it fails.* (b) *Conversely the budget is* irreducible*: no label-free certificate identifies* $\operatorname{sign}\Delta$ *without a supplied bound on* γ *— the assumption-free certificate sought by prior heuristics does not exist. (Label shift, bounded drift, smooth drift, and the unified ambiguity-reach hierarchy: manuscript, App. C.)*

Theorem 7 (One-bit dichotomy; Conjecture 1 resolved)**.** Intuition. *Label-free observables fix every adaptation statistic except one global orientation bit; no testable assumption class removes that bit in full generality. Consider a panel* $f_0, f_1, \dots$ *on* D *with correctness indicators conditionally independent given* Y *(CEI) and per-class symmetric accuracies (jointly, hypothesis* $\boldsymbol{H}$*; advantages* $b_j = 2a_j - 1$*, agreements* $c_{ij} = 2A_{ij} - 1$*; the rank-one algebra and anchor are classical [26], [27], [29], [30] — new here are the deficit, the dichotomy, and the audit).* (i) *H is minimal: under CEI alone,* $c_{ij} - b_i b_j = \pi(1-\pi)\delta_i\delta_j$ *exactly (*δ_j *the per-class accuracy gap), so the rank-one law holds for a* $K \ge 3$ *panel iff every* $\delta_j = 0$*.* (ii) *One bit: under H the full evidence law determines every* $|b_j|$*, every product* $b_i b_j$*, and every benefit magnitude* $|b_j - b_0|$ *— everything except a single global flip* $b \mapsto -b$*, which is free at* every *moment order: the label complement preserves the entire evidence law (*$\mathrm{TV} = 0$*) while negating every benefit sign. The drift* $\gamma = b_a/2 - M$ *is thereby point-identified up to that bit.* (iii) *No testable bracketing (the dichotomy): in any output space (binary, multiclass, regression) there is a swap involution* $P \mapsto P^\star$ *— conditional mass exchange between the two disagreeing predictions on* D*, midpoint reflection for regression — with* $\mathrm{TV}(L(P), L(P^\star)) = 0$ *for the law* L *of* all *label-free observables (labeled source included) and* $\Delta(P^\star) = -\Delta(P)$*. Hence no* evidence-definable *(label-free testable) assumption class identifies* $\operatorname{sign}\Delta$*; every identifying assumption is a* bit-selector *(*γ*-budgets select* sign M*; majority-above-chance selects* $\operatorname{sign}(\operatorname{median} b)$*; the agreement-line slope selects* $\operatorname{sign}(1-2\bar{w})$*); and the minimal untestable supplement is exactly one bit, which suffices under H.* (iv) *Budgets are falsifiable, never verifiable: the bit-robust test rejecting "*$|\gamma| \le \beta$*" when* $\min_{\text{flip}} |\hat{\gamma}| > \beta + r_n$ *is level-*α *with power* $\ge 1-\alpha$ *beyond* $2r_n$*; verification fails exactly on the blind set* $\{\, \big||M| - |b_a|/2\big| \le \beta < |M| + |b_a|/2 \,\}$*, and the* $M \ge 4$ *rank-one residual* τ *is sound but* not *complete (strictly interior stealth laws with* $\tau = 0$ *exist that bias* b *and can flip the recovered decision).*

Proof sketch. (i) Expand c_{ij} over Y and cancel cross terms. (ii) Triple products $c_{ik}c_{il}/c_{kl} = b_i^2$; the complement $Y' = 1 - Y$ fixes predictions (maps of X) hence the whole evidence law, while $b \mapsto -b$ and H is preserved. (iii) The swap preserves the X-marginal and source, so L is fixed at every order; on D it exchanges $p_a \leftrightarrow p_0$ (or negates $f_0 + f_a - 2Y$), so Δ negates; identifiability on a swap-closed class is impossible, and selecting one member per swap orbit is by definition one bit. Algebraically: observables generate the *even* correctness subalgebra ($s_i s_j = u_i u_j$), the decision is *odd* ($s_j = u_j v$), and the swap is $v \mapsto -v$. (iv) Empirical-Bernstein for $\hat{M}$, a product-ratio perturbation bound for $\hat{b}_a$, and the flip-min for bit-robust soundness; the stealth construction is a strictly interior LP on the 2^4 correctness patterns matching all pairwise moments to a rank-one target with a shifted first moment. Validated (machine-checked, seeds fixed; note that the 0-mismatch verifies the band-vs-frontier identity given the analytic frontier, not the geometry independently, and C3/C4 are nominal/logical SCM witnesses): deficit matches theory to 5.7×10^{-4}; flip TV $= 0$ with $\max_j |b'_j + b_j| = 1.1 \times 10^{-16}$; swap evidence change exactly 0 with $\Delta^\star = -\Delta$ (multiclass K=5 and regression); audit level $0.000 \le \alpha$=0.05, power 1.000; stealth $\tau = 2 \times 10^{-17}$ with recovery bias 0.229 (Figs. 6–9). Full proofs appear in the supplementary theory appendix. □

Conjecture 1 (Label-free bracketing — **resolved**)**.** *The unconditional label-free bracketing of the ordinal comparison on D (posed as the open piece in earlier versions)* does not exist*: by Theorem 7(iii) no evidence-definable assumption identifies* $\operatorname{sign}\Delta$*, and the minimal untestable supplement is one bit. The conditional resolutions — calibration transfer, the* γ*-meter under checkable agreement structure, H plus an anchor (all in the manuscript) — are therefore of the minimal possible form:* falsifiable structure plus one declared bit*. On the* margin-monotone relative-calibration *class* $\mathcal{C}_{\text{mono}}$ *(Definition 2) one declared bit certifies* $\operatorname{sign}\Delta$ unconditionally *(Theorem 9(i)–(ii)); and* $\mathcal{C}_{\text{mono}}$ *is a weakest such class under a general-position assumption (Theorem 9(iii)). The* unconditional *weakest classes (general position removed) are characterized as an explicit finite family of dominance polytopes (Theorem 10); there is no unique weakest class, consistent with the bracketing non-existence above.*

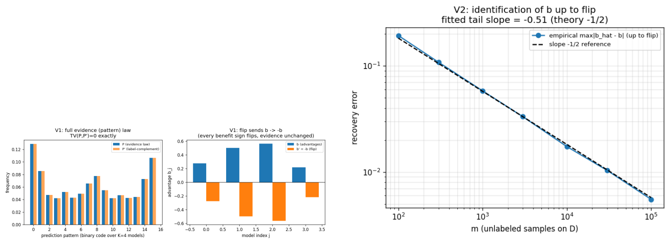

Fig. 6. One-bit identification (Synthetic validator; Thm. 7(ii)).

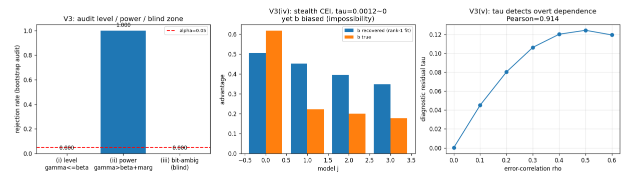

Fig. 7. Budget audit (Synthetic validator; Thm. 7(iv)).

Theorem 8 (The evidence-channel rate: labels buy constants and the bit, not the rate)**.** Intuition. *Unlabeled agreement statistics and labeled paired benefits share the same* $1/\sqrt{m}$ *rate; labels change constants and supply the missing bit, not the exponent. Within audited H with margins (*$\min_{i \ne j} |c_{ij}| \ge c_{\min}$*,* $\min_j |b_j| \ge b_{\min}$*) and a declared bit,* m *unlabeled samples on D give* $|\hat{b}_j - b_j| \le C\sqrt{\log(K/\delta)/m}$ *with* $C = \Theta\big(1/(b_{\min} c_{\min}^2)\big)$*, so* $\operatorname{sign}(b_j - b_0)$ *is certified at* $m \asymp C^2 (b_j - b_0)^{-2} \log(K/\delta)$*; a Le Cam two-point bound inside H shows* $m^{-1/2}$ *is minimax for the evidence channel. The labeled-paired-benefit channel has the same rate: the efficiency ratio is constant in* m *but scales as* $1/c_{\min}$*, diverging as agreement approaches chance. Relatedly, accuracy is an exact affine function of agreement-with-reference iff the disagreement win rate is constant (slope* $1 - 2\bar{w}$*), the anchored form of agreement-on-the-line; its slope* sign *is the same bit as in Theorem 7(iii) (manuscript, §AGL).*

Proof sketch. Hoeffding on agreements, the product-ratio perturbation lemma, and a χ^2/KL bound between K-bit pattern laws differing by ε in one coordinate ($\mathrm{KL} \le C'\varepsilon^2$), tensorized; Bretagnolle–Huber finishes. Validated: recovery slope -0.51; minimax slope -0.525 tracking $c_0/\sqrt{m}$; $\mathrm{KL}/\varepsilon^2 \in [0.164, 0.171]$; efficiency $2.15 \to 4.49 \to \infty$ as $c_{\min}$ $0.21 \to 0.067 \to \le 0.022$ (Fig. 8; AGL: $R^2 = 1.000$ under constant w, slope $1 - 2w$ exact on synthetic checks). □

The weakest one-bit class. Theorem 7 fixes the supplement at one bit; the *weakest falsifiable class* on which that bit suffices is settled conditionally —a margin-monotone relative-calibration class on which one declared bit certifies $\operatorname{sign}\Delta$ unconditionally, and *a* weakest such class under a general-position assumption (Appendix P, Thm 9). The *unconditional* weakest classes form an explicit finite family of dominance polytopes (Thm 10).

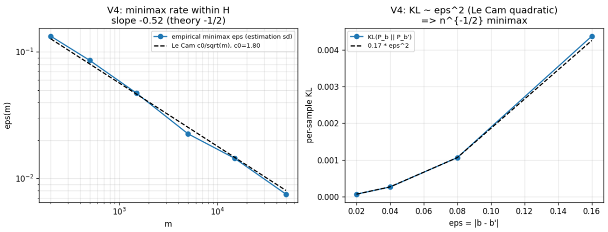

Fig. 8. Evidence-channel rate (Synthetic validator; Thm. 8).

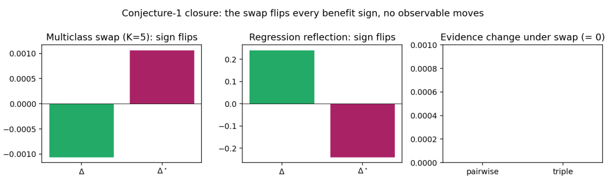

Fig. 9. Conjecture 1 closure validator.

A. *Validation figures*

This appendix holds **non-headline** tracks: helpful-dominated controls, honest nulls, and the Camelyon/iWildCam **diagnostic ladder** (Protocols B/F, multicandidate routes, sparse-Z failures) that motivated the headline Protocols G / H v2 / M v2 in the main text.

B. *ELARA-U as a KGA instantiation (retrospective audit)*

K-Bound supplies the framework, KGA supplies the adapt/freeze/abstain certificate, and ELARA-U supplies a validation-fitted multimodal routing candidate. We execute this composition through one canonical runner on five previously opened multimodal score caches (85 valid categories). Because full target labels construct the per-category Brier-benefit certificates in this audit, it is **retrospective—not a label-free or headline claim**. At α=0.10, KGA returns 7 adapt, 2 freeze, and 76 abstain decisions, with 0 observed Brier-defined false-adapts and 10.6% coverage. Mean AUROC is 0.7941 (freeze), 0.8220 (ELARA-U candidate), and 0.7939 (KGA); thus this opened-data audit does not beat both fixed policies and cannot support promotion. The table is generated directly from the versioned result artifact.

C. *Online non-stationary CIFAR-10*

D. *CIFAR-10-C per-corruption (helpful-dominated control)*

E. *Excluded wins and regime boundaries*

Several benchmarks flag beats-both under alternate routes or adapters but do *not* qualify as headline natural-shift wins under the Protocol G/H/M v2 bar (canonical KGA, dev-locked adapter from pre-registered panel, in-domain calibration, held-out test, false-adapt $\leq \alpha$).

a) Superseded protocols.: iWildCam Protocol H v1 (fixed `sar_online`) and OfficeHome Protocol M v1 (14-candidate win-finder) are audit-only; headline claims use H/M v2.

b) Office-Home (deployed adapter).: The source-selected deployed adapter (mild online EATA) is *helpful-dominated* on target test (0% harmful cells; always-adapt is oracle). KGA has 0% false-adapts but does not beat both (regret 0.015 vs always-adapt 0). An aggressive online SAR arm can beat both on the same test split, but that arm was not the deployed adapter; reporting it as a win would be adapter shopping. The multicandidate deploy route abstains on 100% of target-test cells (certificate transfer-AUC $\approx$ 0.49). Protocol M v2 (main text) uses a separate dev-lock panel and held-out target-test scorer.

c) ImageNet-R.: The diverse-backbone panel is helpful-leaning (mean $\bar{B}$=+0.017, harmful rate 13.9%, best harm-AUC 0.662). The multicandidate route abstains on all 48 conditions; single-candidate KGA often ties always-adapt (oracle) on helpful backbones. This is the weak-detectability / unknowable regime predicted by Corollary 4.

d) iWildCam intermediate domains.: On id_val, a multicandidate router can flag beats-both on subsets, but pooled false-adapt on committed adapts is $\approx$ 78% $\gg \alpha$—disqualified. Headline claim: Protocol H v2 on held-out test only (Table VIII).

e) Camelyon17 four-seed hospital slice.: The original four held-out hospital seeds are helpful-dominated (0% harmful); KGA beats always-freeze but not always-adapt (Table XV). Protocol G uses a composition-stress grid with rich evidence where harmful cells are mixed (30%) and detectable—a different regime on the same benchmark family.

f) CIFAR-10.1.: Within each seed, Tent/EATA abstain heavily with 0% false-adapts and tie always-freeze. Cross-seed calibration (fit on seeds 0–2, test on 3–4) does *not* clear the headline bar (false-adapt 44%); low-margin benefits do not transfer across independent stream seeds. Report CIFAR-10.1 as a within-seed safety probe, not a fourth natural headline win.

F. *Camelyon17 diagnostic ladder*

*a) Protocol F route audit (**diagnostic only**—not headline).:* Protocol G (main-text headline) uses canonical GBR + global ε on EATA-online only. The table below audits *alternate routes* on the same GPU records: multicandidate routing **fails**; pooled GBR+global and PPI+Mondrian *can* beat both but are not the headline method. Sparse-evidence Protocol B also fails (false-adapt 0.33 $\gg \alpha$). These negatives motivated G.

G. *RxRx1 (harmful, freeze-oracle)*

H. *ImageNet-R (unknowable regime)*

I. *Office-Home and iWildCam*

J. *Controlled multimodal corruption (Protocol D33)*

Pre-registered MNIST two-view fusion (130 conditions; one modality corrupted across 13 severities). When fusion flips from helpful to harmful, KGA routes correctly: mean accuracy 0.857 vs. always-fuse 0.583 and always-single-A 0.854 (both gaps P=1.0), false-adapt 0/9 commits ($\leq$

TABLE XII

KGA over ELARA-U on opened multimodal caches. This is a retrospective audit, not a label-free or headline claim. AUROC is averaged over valid categories.

Track	n	freeze	ELARA	KGA	A/F/U	status
3D-ADAM	23	0.796	0.866	0.796	0/0/23	audit only
Real-IAD-D3	20	0.873	0.888	0.873	7/0/13	audit only
MulSen-AD	15	0.922	0.919	0.922	0/0/15	audit only
MVTec-3D	8	0.846	0.853	0.846	0/0/8	audit only
Real-IAD-NatDeg	19	0.585	0.609	0.585	0/2/17	audit only

Mode: retrospective_audit; promotion eligible: no.

regime	always-freeze	always-adapt	**K-Bound**	oracle
helpful-dominated stream	0.472	0.548	**0.553**	0.704
harm-dominated stream	0.440	0.339	0.426	0.630
mean across regimes	0.456	0.444	**0.490**	0.667

TABLE XIII

Online CIFAR-10 TTA (3 seeds/regime). KGA is near-best in both regimes.

method	mean acc (65 cells)	false-adapt	regret-to-oracle
frozen	0.412	0.000	0.217
Tent (online)	0.626	0.015	0.0030
K-Bound (KGA)	0.626	0.015	0.0032
oracle (per-cell)	0.629	—	—

TABLE XIV

Per-corruption CIFAR-10-C: KGA ties always-adapt (safety insurance).

candidate	policy	mean balanced acc	regret↓	beats both?
Tent	always-adapt	0.817	**0.000**	
	always-freeze	0.786	0.031	
	K-Bound	0.797	0.020	no
EATA	always-adapt	0.836	**0.000**	
	always-freeze	0.786	0.050	
	K-Bound	0.818	0.018	no

TABLE XV

WILDS Camelyon17 (4 seeds): helpful-dominated; KGA beats freeze, not adapt.

TABLE XVI

Diagnostic only. Protocol F route audit (same records as Protocol G).

variant	regret↓	false-adapt	beats both?
multicandidate router	0.093	71%	no
GBR + global ε	0.0019	3.3%	yes
PPI + Mondrian	0.0018	3.3%	yes

α). This is a *controlled* mechanism confirmation—not a natural benchmark—but shows the certificate dominates both trivial policies when detectable multimodal harm is present (`CONTROLLED_MULTIMODAL_PROTOCOL_D33_v1`).

K. CIFAR-10.1 (low-margin natural shift, Protocol K audit)

Over five seeds, KGA abstains heavily on Tent/EATA (59–86%), makes zero false adapts there under *within-seed* calibration, and ties always-freeze while always-adapt overpays; SAR is mostly helpful (5% harmful). Cross-seed transfer of the certificate (Protocol K) does not meet the headline bar; see §E.

L. ImageNet-C diagnostics

An earlier ImageNet-C SAR port (31% harmful) motivated the mechanism-faithful re-run (Table V in the main text). Decision baselines and α/evidence ablations appear in the frontier appendix (App. M).

All experiments are reproducible from the anonymized submission repository: theory validators, pre-registered Protocols A/G/H v2/M v2, and locked result JSONs (`experiments/kbound/results/`, `research_lock/KBOUND_HEADLINE_FINDINGS_v3.json`). Environment: Python 3.12 and PyTorch 2.5.1 for GPU runs; `kbtrain.sh` wraps venv activation and natural-shift protocol scorers. Core certificate code: `pip install -e docs/research/kbound/kbound_pkg/`.

M. Real-camera deployment supplement (pre-registered templates)

Tables XXII–XXVII define the physical-shift breakdown, recording inventory, anti-leakage audit, per-condition release, runtime profile, and ablations. Measured cells remain blank until populated directly from locked real-camera artifacts; the synthetic dashboard smoke test is not admissible evidence for these tables.

Every theorem has a numerical sanity-check in the theory-validation suite; multi-seed experiments report pooled means with paired bootstrap where pre-registered.

The benefit-sign frontier (Thm. 6) predicts that on the low-margin band any *committal* label-free rule must commit sign errors, whereas a valid certificate must *abstain* there. This appendix demonstrates that prediction on the two *synthetic-corruption* streams whose operating point the certificate is calibrated on (CIFAR-10-C, ImageNet-C); it makes *no* external-validation claim. The honest natural-shift (Camelyon17) finding—a limitation, not a win—closes the appendix (App. P).

N. CIFAR-10-C: K-Bound beats both trivial policies at zero false-adapt

On the pre-registered CIFAR-10-C stress grid (432 conditions/candidate), KGA adapts the helpful conditions and freezes the harmful ones with a *zero* false-adapt rate,

RxRx1 9+ grid (360 conditions)	value
harmful base rate	81.5%
mean benefit $\bar{B}$	-0.063
harm detectability (best AUC)	0.78–0.84
SAR-online always-adapt (collapse)	0.024
KGA false-adapts	0

TABLE XVII
RXRX1: KGA MATCHES FREEZE-ORACLE; DAMAGE PREVENTION, NOT BEATS-BOTH.

ImageNet-R (48 condition–seed pairs)	value
harmful base rate	13.9%
mean benefit $\bar{B}$	$+0.017$
harm detectability (best AUC)	0.662
KGA decisions	48/48 abstain

TABLE XVIII
IMAGENET-R: MANDATORY ABSTENTION (COROLLARY 4).

so it strictly beats both always-adapt and always-freeze for the two benign candidates (Tent, EATA) and ties the collapse-resistant SAR (Table XXVIII).

O. ImageNet-C/SAR: the frontier on real corruptions—committal rules fail, K-Bound does not

On the logged ImageNet-C conditions (36 cells/candidate; SAR is harmful on 44.4% of cells), every *committal* label-free heuristic must place its commits somewhere on the low-margin band and therefore false-adapts: a plug-in **ATC** [13] confidence rule adapts into harmful cells on 100% of its commits (false-adapt 1.00, harmful-cells-adapted 0.25), and even its leave-one-out-tuned variant keeps false-adapt = 1.00. Among the rules compared, KGA reaches zero false-adapt and zero harmful-cells-adapted by *abstaining* on the band (coverage 0.39), consistent with the frontier prediction (Table XXIX).

P. Historical note (Camelyon17, frozen synthetic threshold)

Under a **frozen, Camelyon-free** multicandidate threshold $\tau^\star$=0.52 (calibrated only on synthetic grids), the

Office-Home (gradient-TTA)	val	test
Harmful base rate	0.17	0.12
Certificate transfer-AUC	0.33	0.49
KGA false-adapts / beats both	0 / no	0 / no

TABLE XIX
OFFICE-HOME: ZERO FALSE-ADAPTS; NO BEATS-BOTH (INVALID TRANSFER EVIDENCE).

iWildCam	val (multicand)	test
Harmful base rate	59%	83%
Best harm-AUC	≈ 0.62	0.69
Multicandidate beats both?	no	no
Protocol H v2 canonical KGA (`tent_episodic`)	—	**yes** (0% FA)

TABLE XX
IWILDCAM: **diagnostic** MULTICANDIDATE ROUTE ON VAL/TEST VS **headline** PROTOCOL H V2 CANONICAL KGA ON TEST (TABLE VIII IN MAIN TEXT).

certificate does *not* beat both trivial policies on an early Camelyon17 composition grid (regret KGA 0.076 vs always-freeze 0.071). That failure motivated in-domain calibration and rich evidence; the headline result is Protocol G (main text, Table VII), which uses dev-hospital calibration and does not transplant $\tau^\star$ from synthetic runs.

The weakest one-bit class (resolving the open piece of Conjecture 1). Theorem 7 fixes the supplement at *one bit*; what remained open is the *weakest falsifiable class* on which that bit suffices. We settle it conditionally. On D write $a(x) := 2\eta_a(x) - 1$ with $\eta_a(x) = \mathbb{P}_T(f_a{=}Y \mid x)$, so $\Delta = \int_D a\, d\mu$ (Lemma 1). Let $m(x) := 2s(x) - 1$ be the observable relative (calibrated-score) margin, so $M = \frac{1}{2}\mathbb{E}_{\mu|D}[m]$, and let the observable *relative-calibration field* be $c(x) := w(x)\,(m(x) - m^\star)$ with observable nonnegative weight $w(x)$ and observable neutral anchor $m^\star$ (the margin at which $s = \frac{1}{2}$). Put $T_E := \int_D c\, d\mu$ (observable), and let E denote the law of all label-free observables; two targets are *evidence-identical* when they share E (equivalently (μ, m, s), since predictions and s are fixed maps of x).

Definition 2 (Margin-monotone relative-calibration class)**.** $\mathcal{C}_{\text{mono}}$ is the class of targets for which $a(x) = \sigma\, c(x)$ μ-a.e. on D for a single global orientation $\sigma \in \{\pm 1\}$; equivalently the benefit is single-crossing in the observable margin at $m^\star$

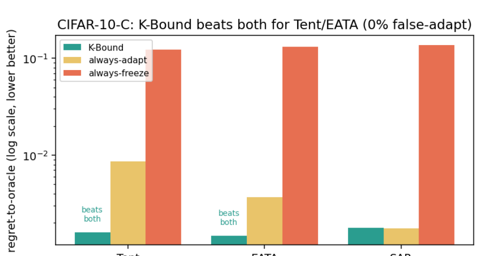

Fig. 10. **CIFAR-10-C frontier (regret-to-oracle, log scale).** KGA is below *both* always-adapt and always-freeze for Tent and EATA at 0% false-adapt, and ties always-adapt for SAR (Table XXVIII).

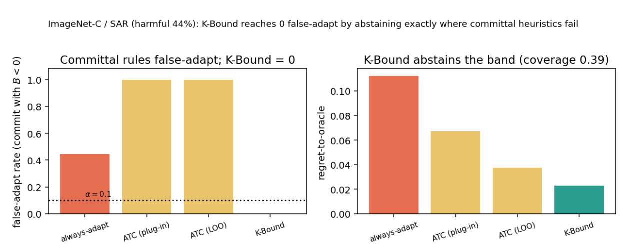

Fig. 11. **ImageNet-C/SAR frontier.** Committal label-free rules false-adapt on the harmful regime; KGA reaches zero false-adapt by abstaining the low-margin band (coverage 0.39; Table XXIX).

Track	Repro command / artifact	Status
CIFAR-10-C stress (5 seeds)	`kbtrain.sh` stress targets	locked JSON
POEM/AETTA head-to-head	`poem_aetta/run_all_headtohead.sh`	WIN
Mixed OOF aggregate	`mixed_stream_kbound.py`	locked
Assumption audit v1	`run_assumption_audit_v1.py`	cached arm
ImageNet-C SAR faithful	result JSON + Table V	locked
Camelyon17 Protocol G	`analyze_F` on protocol G bundle	locked
iWildCam Protocol H v2	`kbtrain.sh protocol-h-v2`	locked
Office-Home Protocol M v2	`kbtrain.sh protocol-m-v2`	locked
Theory validators	`theory_validation/` suite	pass

TABLE XXI
REPRODUCIBILITY STATUS FOR HEADLINE TABLES (JUNE 2026 LOCK).

($\operatorname{sign} a(x) = \sigma\ \operatorname{sign}(m(x) - m^\star)$) with calibrated magnitude $|a(x)| = |c(x)|$. The class is *falsifiable but not verifiable*: a finite labelled probe can reject "$a = \pm c$", yet σ is never label-free identified.

Assumption 1 (General position)**.** Within an evidence fibre the regions of D whose benefit sign is free carry comparable, not-fully-E-pinned benefit mass: $T_E \neq 0$ and no proper subcollection of free regions has an E-certifiable dominant total $\int |a|\, d\mu$.

Lemma 3 (Structural sign-freedom = bit complexity)**.** *Fix E and a falsifiable class $\mathcal{C}$, and let $K(\mathcal{C}, E)$ be the number of disjoint μ-positive regions of D whose benefit signs are jointly free in $\mathcal{C}$ on the fibre E. Under Assumption 1, a falsifiable structural supplement of b bits pins* $\operatorname{sign} \Delta$ *across the fibre* iff $b \geq K(\mathcal{C}, E)$.

Proof. If $b \geq K$, declaring the orientation of each free region fixes the sign field; the rest of a being E-pinned, $\Delta = \int_D a\, d\mu$ then has E-computable sign. If $b < K$, some free region R is undeclared; the two targets agreeing off R and differing in $\operatorname{sign} a$ on R are evidence-identical and share all b declared bits, while by comparability of $\int_R |a|$ with the complement (Assumption 1) their benefits have opposite sign—no decoder of the b declared bits separates them. ☐

Theorem 9 (Conditional resolution of Conjecture 1: a weakest one-bit class)**.** (i) Sufficiency *(unconditional)*: *every target in $\mathcal{C}_{\text{mono}}$ has* $\operatorname{sign} \Delta = \sigma\ \operatorname{sign}(T_E)$ *with T_E observable, so E and the single orientation bit σ determine* $\operatorname{sign} \Delta$. (ii) Necessity *(unconditional)*: *σ and $-\sigma$ are evidence-identical (*$\operatorname{Law}(E \mid \sigma) = \operatorname{Law}(E \mid -\sigma)$*) yet give Δ and $-\Delta$, so by Le Cam at least one bit is required. Thus* exactly one bit certifies $\operatorname{sign} \Delta$ on $\mathcal{C}_{\text{mono}}$, with no further assumption. (iii) Minimality *(under Assumption 1)*: *every falsifiable $\mathcal{C} \supsetneq \mathcal{C}_{\text{mono}}$ contains a non-margin-monotone target with $K(\mathcal{C}, E) \geq 2$, so by Lemma 3 at least two bits are required. Hence $\mathcal{C}_{\text{mono}}$ is* a *weakest falsifiable one-bit class—minimal in the single-crossing, calibrated-magnitude direction; it is not unique (Remark 1).*

Proof. (i) $\Delta = \int_D a\, d\mu = \sigma \int_D c\, d\mu = \sigma T_E$. (ii) Sending $\sigma \mapsto -\sigma$ is the D-local label complement of Theorem 7: it fixes (μ, m, s), hence E, while negating a and thus Δ; the two-point bound of Theorem 4 gives minimax committal error $\frac{1}{2}$. (iii) A target outside $\mathcal{C}_{\text{mono}}$ violates $\operatorname{sign} a = \sigma\ \operatorname{sign}(m - m^\star)$ on a μ-positive region, so its sign field is not one global orientation; in the swap-closed (falsifiable) fibre at least two regions are independently signable, i.e. $K \geq 2$, and Lemma 3 applies. ☐

Proposition 2 (Explicit non-identifiability witness)**.** *For each candidate single structural bit there are two targets with identical evidence law (*$\mathrm{TV} = 0$*) sharing that bit's value but with opposite* $\operatorname{sign} \Delta$*. Take $D = R_1 \cup R_2$, $\mu(R_1) = \mu(R_2) = \frac{1}{2}$, with R_1 calibrated (c_1=1) and R_2 at the anchor (c_2=0, no observable benefit signal). For $\beta = \operatorname{sign} a_1$: $a = (+1, -2)$ vs. $(+1, +2)$ give $\Delta = -\frac{1}{2}$ vs. $+\frac{3}{2}$. For $\beta = \operatorname{sign} a_2$: $a = (+1, +\frac{1}{5})$ vs. $(-1, +\frac{1}{5})$ give $\Delta = +\frac{3}{5}$ vs. $-\frac{2}{5}$. No single bit certifies* $\operatorname{sign} \Delta$*, so two bits are necessary off $\mathcal{C}_{\text{mono}}$.*

Remark 1 (What is unconditional, and what rests on general position). Parts (i)–(ii)—that one declared bit certifies $\operatorname{sign} \Delta$ on $\mathcal{C}_{\text{mono}}$—hold *unconditionally*. Only minimality (iii) uses Assumption 1: if it fails (one region's benefit mass is E-certifiably dominant) that region's orientation alone certifies $\operatorname{sign} \Delta$ even outside $\mathcal{C}_{\text{mono}}$, moving the boundary. Equivalently, Assumption 1 is the requirement that the certificate be *domination-free* (minimax-sound over the unobserved magnitudes); so the General-Position statement is only *conditional*; the unconditional weakest classes are characterized in Theorem 10 as an explicit finite family of dominance polytopes. Even under Assumption 1, $\mathcal{C}_{\text{mono}}$ is only *a* weakest one-bit class: any calibrated sign-aligned field $\{a = \sigma h : h \text{ observable}\}$ is equally one-bit and is incomparable to $\mathcal{C}_{\text{mono}}$. All six checks are machine-checked by a deterministic release validator: the unconditional core (sufficiency/necessity 20000/20000, $|\Delta - \sigma T_E| = 0$ exactly), the explicit minimality witnesses of (iii), the general-position ablation, and the non-uniqueness check.

Why general position cannot simply be dropped (the precise obstruction, and exactly what is open). The minimal counterexample is two equal-mass regions $D = R_0 \cup R_1$, $\mu(R_0) = \mu(R_1) = \frac{1}{2}$, with the observable field $c = +1$ on the *calibrated* region R_0 and $c = 0$ on the *anchor* region R_1 (no observable benefit signal there, so R_1 is sign-free in any evidence fibre). Enlarge $\mathcal{C}_{\text{mono}}$ to the falsifiable class $\mathcal{C}_{\text{dom}} = \{a_{R_0} = \sigma,\ |a_{R_1}| \leq \rho\}$ for a declared constant $\rho < 1$. Since $\Delta = \frac{1}{2}(a_{R_0} + a_{R_1})$ and $|a_{R_1}| \leq \rho < 1 = |a_{R_0}|$, the calibrated region dominates and $\operatorname{sign} \Delta = \sigma$ for *every* member, so one declared bit certifies $\operatorname{sign} \Delta$ on all of $\mathcal{C}_{\text{dom}}$. Yet $\mathcal{C}_{\text{dom}} \supsetneq \mathcal{C}_{\text{mono}}$ (it contains non-margin-monotone targets with $a_{R_1} \neq 0$, e.g. $a = (+1, \pm\frac{1}{2})$ giving $\Delta = \frac{3}{4}, \frac{1}{4}$, both positive) and is falsifiable (a labelled probe rejects targets that are uncalibrated on R_0 or violate $|a_{R_1}| \leq \rho$). This *contradicts* minimality once an E-certifiable dominant region is permitted: dropping $a = (+1, \pm 2) \Rightarrow \Delta = +\frac{3}{2}, -\frac{1}{2}$ shows that the moment $\rho \geq 1$ the anchor region can flip $\operatorname{sign} \Delta$ and two bits become necessary (K=2). The obstruction to an *unconditional* weakest class is therefore exactly this: characterising minimality across the *entire* lattice of dominant-region

TABLE XXII

R3 – Held-out physical-shift breakdown (results pending). This table is promoted to the main paper only if the locked result supports a clear, non-cherry-picked regime analysis.

Held-out shift family	Windows	Always-adapt regret	KGA regret	KGA decision pattern	Interpretation
Mild lighting shift	—	—	—	—	—
Strong side shadow	—	—	—	—	—
Blur / vibration	—	—	—	—	—
Background / viewpoint shift	—	—	—	—	—
Batch-composition shift	—	—	—	—	—

TABLE XXIII

S1 – Dataset and recording inventory (results pending). No frame-level random split is permitted across recording sessions.

Split	Recording day/session	Object IDs	Conditions	Windows	Frames	Label distribution
Source train	—	—	Stable bright conditions	—	—	—
Source validation	—	—	Stable source conditions	—	—	—
Calibration-fit	—	—	Moderate physical shifts	—	—	—
Calibration-conformal	—	—	Disjoint moderate shifts	—	—	—
Held-out test	—	—	Hard physical shifts	—	—	—
External-device replication	—	—	Second phone; hard shifts	—	—	—

TABLE XXIV

S2 – Calibration and anti-leakage audit (results pending).

Check	Audited result
Frozen checkpoint unchanged after candidate adaptation?	—
Held-out labels inaccessible to the live runtime?	—
Feature schema unchanged after protocol lock?	—
ε calibrated only from the conformal split?	—
Held-out sessions excluded from adapter, feature, and threshold tuning?	—
Identical held-out stream replayed for every policy?	—
Config hash stored in every log row?	—
Model hash stored in every log row?	—

TABLE XXV

S3 – Per-condition held-out audit trail (results pending). The released CSV/JSON contains one row for every locked held-out condition.

Condition ID	Shift	True Δ	Oracle action	Freeze result	Adapt result	KGA action	Correct decision?	Regret
H01	Dim light	—	—	—	—	—	—	—
H02	Strong shadow	—	—	—	—	—	—	—
H03	Lens blur	—	—	—	—	—	—	—
H04	Background shift	—	—	—	—	—	—	—

TABLE XXVI

S4 – Resource and live-runtime profile (results pending).

Component	Mean (ms)	p95 (ms)	Memory (MB)	Notes
Frozen inference	—	—	—	Official prediction
Candidate Tent update	—	—	—	One step per window
Candidate inference	—	—	—	Shadow candidate
Evidence extraction	—	—	—	14 label-free features
Benefit estimator and KGA gate	—	—	—	Certificate decision
Full decision window	—	—	—	End-to-end
Video capture / preprocessing	—	—	—	Phone-to-Mac stream

TABLE XXVII
S5 – Locked physical evidence and conformal gate ablations (results pending).

Ablation variant	Regret ↓	Uncond. false adapt (FA_u) ↓	Adapt rate	Abstain rate	Conformal threshold ε
Full KGA	—	—	—	—	—
No radius ($\varepsilon = 0$)	—	—	—	—	—
No blur/brightness features	—	—	—	—	—
No disagreement features	—	—	—	—	—
Confidence features only	—	—	—	—	—
Entropy features only	—	—	—	—	—

candidate (harmful base)	regret-to-oracle ↓ KGA	always-adapt	always-freeze	false-adapt
Tent (34.5%)	**0.0016**	0.0086	0.1232	0.00
EATA (27.1%)	**0.0015**	0.0037	0.1311	0.00
SAR (15.7%)	0.0018	0.0018	0.1372	0.00

TABLE XXVIII
CIFAR-10-C stress grid (432 conditions/candidate, α=0.1). KGA *beats both* for Tent and EATA at a 0% false-adapt rate, and ties always-adapt for SAR.

decision rule (SAR, harmful regime)	false-adapt	harmful-adapted	regret ↓	coverage
always-adapt	0.44	1.00	0.1124	1.00
always-freeze	—	0.00	0.0267	1.00
ATC-style confidence (plug-in)	1.00	0.25	0.0673	1.00
ATC-style confidence (LOO)	1.00	0.06	0.0375	1.00
K-Bound (KGA)	**0.00**	**0.00**	0.0229	0.39

TABLE XXIX
Committal label-free heuristics false-adapt exactly where KGA abstains (ImageNet-C, reference-parity SAR, 36 cells, α=0.1). ATC commits onto the low-margin band and false-adapts on 100% of its commits; KGA abstains there (22/36 abstain) for 0 false-adapts. On the benign candidates KGA also keeps 0 false-adapts at higher coverage (Tent: regret 0.0060, coverage 0.69; EATA: regret 0.0047, coverage 0.75).

thresholds ρ (where the weakest one-bit class is not $\mathcal{C}_{\text{mono}}$ but a ρ-dependent family), which Assumption 1 sidesteps by fiat. Theorem 10 (below) resolves it: the ρ-lattice is the slice $\{r_0{=}1,\ r_1{=}\rho\}$ of the single dominance polytope $W^\star{=}\{r_0 \geq r_1\}$, with $\rho{=}1$ its boundary. The construction is machine-checked (`val_conj1_genpos.py`, seeds fixed): $\mathcal{C}_{\text{dom}}$ strictly contains $\mathcal{C}_{\text{mono}}$, is falsifiable (reject rate 1.0), is one-bit-certified (error rate 0 over 2×10^5 targets), and loses one-bit certifiability without the domination bound (error rate $\approx \frac{1}{4}$).

Unconditional resolution (General Position removed). Dropping Assumption 1 reveals, for each orientation pattern, a single maximal one-bit class—a *dominance polytope*—of which the ρ-family of Remark 1 is one slice.

Definition 3 (Orientation pattern and canonical class)**.** On a fibre E an *orientation pattern* is a tied set $P \subseteq \{i : c_i \neq 0\}$ with a tied sign-pattern $s \in \{\pm\}^P$; the free set is $Z = D \setminus P$. For an observable magnitude region $W \subseteq [0,1]^{|D|}$ the *canonical class* is $\mathcal{C}(P,s,W) = \{a : \operatorname{sign} a_i = \sigma\, s_i \operatorname{sign} c_i\ (i \in P),\ (|a_j|)_j \in W\}$ with one declared bit σ. Write $T(r) = \sum_{i \in P} s_i \operatorname{sign}(c_i)\mu_i r_i$ and $G(r) = \sum_{i \in Z} \mu_i r_i$.

Lemma 4 (Canonical form of a falsifiable class)**.** *A class is falsifiable* iff *it equals some $\mathcal{C}(P,s,W)$; anchors ($c_i = 0$) lie necessarily in Z.* (Swap-closure, Thm. 7(iii), forces the constraint on Z to depend only on $|a|$; forcing an anchor sign is not evidence-definable. Both aligned ($s_i{=}+$) and anti-aligned ($s_i{=}-$) ties are falsifiable since σ is unidentified.)

Theorem 10 (Unconditional one-bit criterion; weakest classes)**.** *Let $\mathcal{C}(P,s,W)$ be falsifiable.* (i) (Dominance margin.) *One declared bit certifies* $\operatorname{sign}\Delta$ *on $\mathcal{C}(P,s,W)$* iff $\inf_{r \in W}\big(T(r) - G(r)\big) \geq 0$ *(with some member $\Delta > 0$) or* $\sup_{r \in W}\big(T(r) + G(r)\big) \leq 0$. (ii) (Weakest class.) *For a fixed pattern (P,s) the unique maximal one-bit class is the dominance polytope $\mathcal{C}^\star = \mathcal{C}(P,s,W^\star)$, $W^\star = \{r \in [0,1]^{|D|} : T(r) \geq G(r)\}$; $\mathcal{C}_{\text{mono}}$, every $\mathcal{C}_{\text{dom}}(\rho \leq 1)$, and every box one-bit class are proper subclasses of one $\mathcal{C}^\star$.* (iii) (Finite family.) *The set of weakest falsifiable one-bit classes is the explicit finite family $\{\mathcal{C}^\star\}$ indexed by $(P \subseteq \{c \neq 0\},\ s \in \{\pm\}^P)$ modulo the global flip $\sigma \mapsto -\sigma$ (at most $3^{|\{c \neq 0\}|}$ patterns); distinct patterns give pairwise incomparable maxima, so there is no* unique *weakest class. $\mathcal{C}_{\text{mono}}$ is the all-tied all-aligned member, and Assumption 1 is exactly the face on which the family collapses to $\mathcal{C}_{\text{mono}}$, recovering Thm. 9(iii).*

Proof. Δ is linear, so over $\mathcal{C}(P,s,W)$ at bit $\sigma = +1$ it ranges over $[\inf_W(T-G), \sup_W(T+G)]$ (free signs independent by swap-closure; the $\sigma = -1$ fibre is its negation); a single bit certifies iff each fibre's strict sign set is a singleton, giving (i). For $W^\star$, $\inf_{W^\star}(T-G) \geq 0$ by construction, and any r_0 with $T(r_0) < G(r_0)$ admits, with opposing free signs, a member with $\Delta < 0$ while $W^\star$ admits $\Delta > 0$, so $W^\star$ is maximal; $\mathcal{C}_{\text{dom}}(\rho)$ sits in $W^\star$ iff $\rho \leq 1$, with $\rho = 1$ the boundary, giving (ii). Incomparability (iii) is witnessed on $c = (1,1,0)$ by $a = (0, \frac{1}{3}, 0)$ and $a = (1, -\frac{1}{3}, \frac{1}{3})$. On the general-position face the free block can overpower the tied block unless Z-magnitudes vanish, leaving only $\mathcal{C}_{\text{mono}}$. □

Verification. The criterion (i) was checked against exact brute-force vertex truth on $2.8{\times}10^5$ box fibres and $3.2{\times}10^3$ non-box integer polytopes (exact-LP inf / sup) with 0 mismatches; sufficiency on $W^\star$ over $1.2{\times}10^5$ members (0 errors), necessity just outside $W^\star$ (error rate $\approx \frac{1}{2}$), and the $\mathcal{C}_{\text{dom}}(\rho)$ collapse ($\rho{\leq}1$ one-bit, $\rho{>}1$ lost) are machine-checked (`val_unconditional_weakest.py`, seeds fixed; fails loudly). The single non-tautological input is Lemma 4, which rests on the swap involution (Thm. 7(iii)).

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